9 Circular measure and trigonometric functions

In this chapter you will learn:

- about different units for measuring angles, and how measuring angles is related to distance travelled around a circle
- the definitions of sine, cosine and tangent functions, their basic properties and their graphs
- how to calculate certain special values of trigonometric functions
- how to apply your knowledge of transformations of graphs to sketch more complicated trigonometric functions
- how to use trigonometric functions to model periodic phenomena
- about inverses of trigonometric functions.

Introductory problem

The original Ferris Wheel was constructed in 1893 in Chicago. It was just over 80 m tall and could complete one full revolution in 9 minutes. During each revolution, how long did the passengers spend more than 5 m above ground?

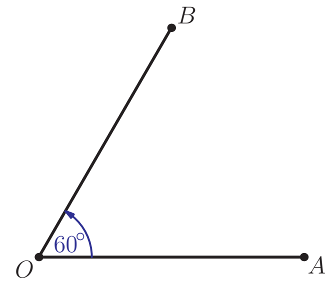
Measuring angles is related to measuring lengths around the perimeter of the circle. This observation leads to the introduction of a new unit for measuring angles, the **radian**, which will be a very useful unit of measurement in advanced mathematics.

Motion in a circle is just one example of periodic motion, which repeats after a fixed time interval. Other examples include oscillation of a particle attached to the end of a spring or vibration of a guitar string. There are also periodic phenomena where a pattern is repeated in space rather than in time; for example, the shape of a water wave. All these can be modelled using **trigonometric functions**.

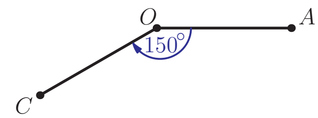
9A Radian measure

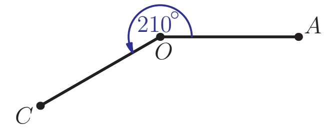
An angle measures the amount of rotation between two straight lines. You are already familiar with measuring angles in **degrees**, where a full turn measures 360° and that there are two directions (or senses) of rotation; clockwise and anti-clockwise. It is conventional in mathematics to measure angles anti-clockwise.

In the first diagram alongside, the line OA rotates 60° anti-clockwise into position OB. Therefore $A\hat{O}B = 60°$.



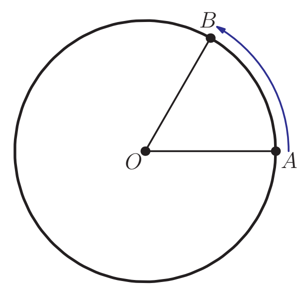
In the second diagram the line OA rotates 150° clockwise into position OC. We measure clockwise rotations with negative angles. Therefore $A\hat{O}C = -150°$. Note that OA can also move to position OC by rotating 210° anti-clockwise, so also $A\hat{O}C = 210°$, as shown in the third diagram. In this book, and in exam questions, it will always be made clear which of the two angles is required.





The measure of 360° for a full turn may seem arbitrary and there are other ways of measuring sizes of angles. In advanced mathematics, the most useful unit for measuring angles is the **radian**. This measure relates the size of the angle to the distance moved by a point around a circle.

Consider a circle with centre O and radius 1 (this is the **unit circle**), and two points, A and B, on its circumference. As the line OA rotates into position OB, point A moves the distance equal to the length of the arc AB. The measure of the angle AOB in radians is equal to this arc length.



If point A makes a full rotation around the circle, it will cover a distance equal to the length of the circumference of the circle. As the radius of the circle is 1, the length of the circumference is 2π. Hence a full turn measures 2π radians. We can deduce the sizes of other common angles in radians; for example, a right angle is one quarter of a full turn, so it measures $2\pi \div 4 = \frac{\pi}{2}$ radians.

Common angles, measured in radians, are often expressed as fractions of π but we can also use decimal approximations. Thus a right angle measures approximately 1.57 radians.

The fact that a full turn measures 2π radians can be used to convert any angle measurement from degrees to radians, and vice versa.

You may wonder why there are 360 degrees in a full turn. It came from ancient astrologers who noticed that the stars appeared to rotate in the sky, returning to their original position after around 360 days. One day's rotation corresponds to one degree. We now know that there are 365.24 days in a year, so these ancient astrologers were quite accurate – you might want to think how you could measure how many days there are in a year.

Worked example 9.1

(a) Convert 75° to radians. (b) Convert 2.5 radians to degrees.

What fraction of a full turn is 75°?

(a) $\frac{75}{360} = \frac{5}{24}$

Calculate the same fraction of 2π

$\frac{5}{24} \times 2\pi = \frac{5\pi}{12}$

This is the *exact* answer. We can also find the decimal equivalent, to 3 significant figures.

$\therefore 75° = \frac{5\pi}{12}$ radians

$75° = 1.31$ radians (3SF)

What fraction of a full turn is 2.5 radians?

(b) $\frac{2.5}{2\pi}$ (≈ 0.3979)

Calculate the same fraction of 360°

$\frac{2.5}{2\pi} \times 360 = 143.24\ldots$

2.5 radians = 143° (3SF)

KEY POINT 9.1

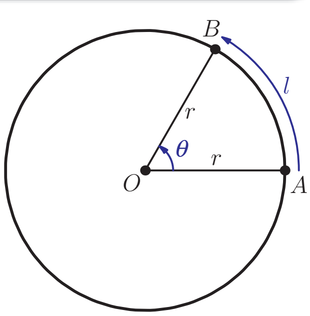
Full turn = 360° = 2π radians

Half turn = 180° = π radians

To convert from degrees to radians, divide by 180 and multiply by π.

To convert from radians to degrees, divide by π and multiply by 180.

There are many different measures of angle. One historical attempt was gradians, which split a right angle into 100 units. Does this mean that the facts you have learnt about angles – ideas such as 180° in a triangle – are purely consequences of definitions and have no link to truth?



In our definition of the radian measure we have used the unit circle. However, we can think about a point moving around a circle of any radius.

As the line OA rotates into position OB, point A covers the distance equal to the length of the arc AB. The ratio of this arc length l to the circumference of the circle is $\frac{l}{2\pi r}$.

The measure of $A\hat{O}B$ in radians is in the same ratio to the full turn, so $\frac{\theta}{2\pi} = \frac{l}{2\pi r}$.

It follows that the measure of the angle in radians is $\theta = \frac{l}{r}$; the measure of the angle is the ratio of the length of the arc to the radius of the circle. In particular, the angle of 1 radian corresponds to an arc whose length is equal to the radius of the circle.

When the radius of the circle is 1, the size of the angle is numerically equal to the length of the arc. However, as the size of the angle is a ratio of two lengths, it has no units. We say that the radian is a *dimensionless unit*. For this reason, when writing the size of an angle in radians, we will simply write, for example, $\theta = 1.31$.

All this may sound a little complicated and you may wonder why we cannot just use degrees to measure angles. You will see in chapter 11 that the formulae for calculating lengths and areas for parts of circles are much simpler when radians are used, but an even stronger case for using radians will be presented when you study calculus.

One advantage of thinking of angles as measuring the amount of rotation around the unit circle is that we can show an angle of any size by marking the corresponding point on the unit circle. We have seen that the convention in mathematics is to measure positive angles anti-clockwise. Another convention is that we start measuring from a point which is on the positive x-axis. On the diagram alongside, the starting point is labelled A, and the point P corresponds to the angle of 60°. In other words, to get from the starting point to point P, we need to rotate 60°, or one-sixth of the full turn, around the circle.

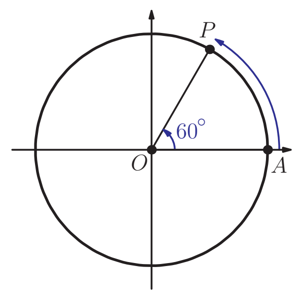
We can also represent negative angles (by rotating clockwise) and angles larger than 360° (by rotating through more than a full turn). The second diagram shows point P representing the angle of 450°, or one and a quarter turns.

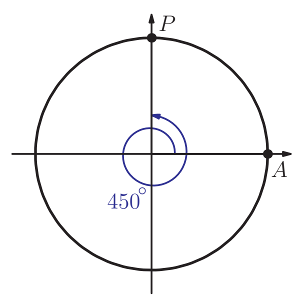
We can apply this idea to representing numbers; instead of representing numbers by points on a number line, we can represent them by points on the unit circle. To do this, we imagine wrapping the number line around a circle of radius 1, starting by placing zero on the positive x-axis (on the diagram on the next page, point S corresponds to number 0), and going anti-clockwise. As the length of the circumference of the circle is 2π, the numbers 2π, 4π, and so on are also represented by point S. Numbers π, 3π, and so on are represented by point P. Number 3, which is just less than π, is represented by point A as shown in the diagram on the next page.

EXAM HINT

Some books write $\theta = 1.31$ rad or $\theta = 1.31^c$, but the IB® does not use this notation.

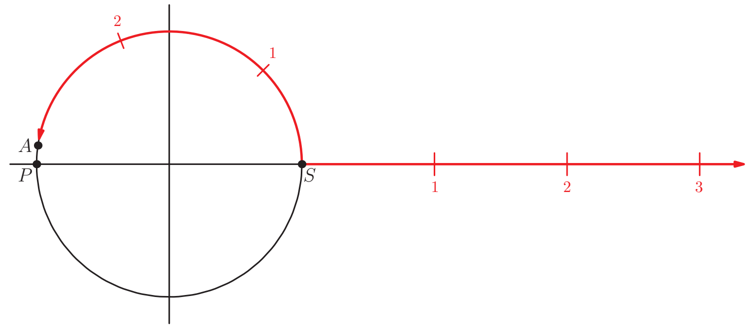
Radians will be used whenever differentiating and integrating trigonometric functions (chapters 16 to 20).



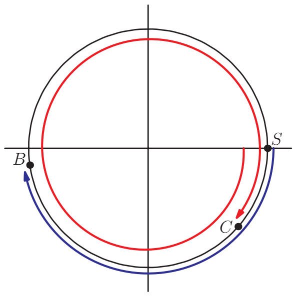


There is a three-dimensional analogue of angle called solid angle, the unit of which is steradians. This unit measures the fraction of the surface area of a sphere covered. There are many aspects of trigonometry which can be transferred to these solid angles.

Looking at the number line in this way should also suggest why radians are a more natural measure of angle than degrees. While the radian is rarely used outside of the mathematical community, within mathematics it is the unit of choice.



We can also represent negative numbers in this way, by wrapping the negative part of the number line clockwise around the circle. For example, number −3 is represented by point B on the diagram alongside and number −7 by point C (7 is a bit bigger than 2π, which means wrapping once and a bit around the circle).



Worked example 9.2

(a) Mark on the unit circle the points corresponding to the following angles, measured in degrees:

A: 135° B: 270° C: −120° D: 765°

(b) Mark on the unit circle the points corresponding to the following angles, measured in radians:

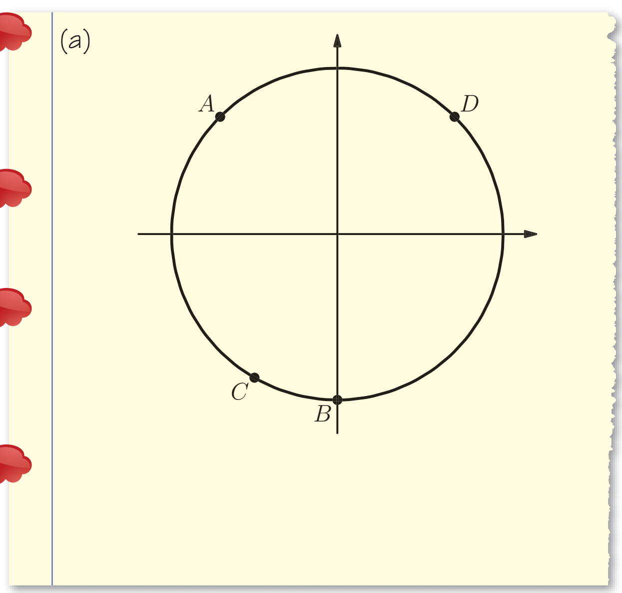
A: π B: $-\dfrac{\pi}{2}$ C: $\dfrac{5\pi}{2}$ D: $\dfrac{13\pi}{3}$

135 = 90 + 45, so point A represents quarter of a turn plus another eighth of a turn

270 = 3 × 90, so point B represents rotation through three right angles

120 = 360 ÷ 3, so point C represents a third of the full turn, but clockwise (because of the minus sign)

765 = 2 × 360 + 45, so point D represents two full turns plus one half of a right angle



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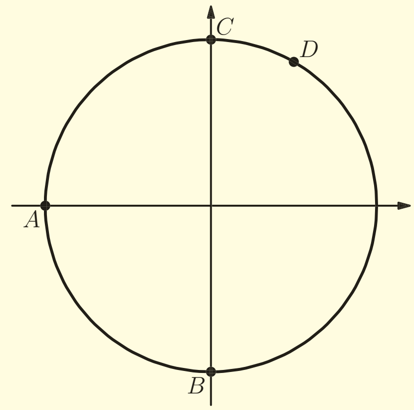
π radians is one half of a full turn so point A represents half a turn

$\frac{\pi}{2}$ is one quarter of a full turn, so point B represents a quarter of a turn in the clockwise direction (because of the minus sign)

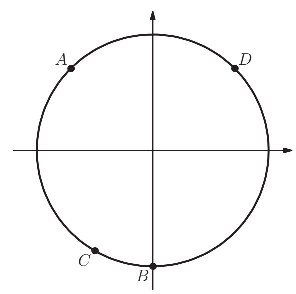
$\frac{5\pi}{2} = 2\pi + \frac{\pi}{2}$, so point C represents a full turn followed by another quarter of a turn

$\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$, so point D represents two full turns followed by another 6th of a turn

(b)



It is useful to describe where points are on the unit circle using quadrants. A **quadrant** is one quarter of the circle, and conventionally quadrants are labelled going anti-clockwise. So in the example (a) above, point A is in the second quadrant, point C in the third quadrant and point D in the first quadrant.



Exercise 9A

1. Draw a unit circle for each part and mark the points corresponding to the following angles.

(a) (i) 60° (ii) 150°

(b) (i) –120° (ii) –90°

(c) (i) 495° (ii) 390°

2. Draw a unit circle for each part and mark the points corresponding to the following angles.

(a) (i) $\frac{\pi}{4}$ (ii) $\frac{\pi}{3}$

(b) (i) $\frac{4\pi}{3}$ (ii) $\frac{3\pi}{4}$

(c) (i) $-\frac{\pi}{3}$ (ii) $-\frac{\pi}{6}$

(d) (i) -2π (ii) -4π

3. Express the following angles in radians, giving your answer in terms of π.

(a) (i) 135° (ii) 45°

(b) (i) 90° (ii) 270°

(c) (i) 120° (ii) 150°

(d) (i) 50° (ii) 80°

4. Express the following angles in radians, correct to 3 decimal places.

(a) (i) 320° (ii) 20°

(b) (i) 270° (ii) 90°

(c) (i) 65° (ii) 145°

(d) (i) 100° (ii) 83°

5. Express the following angles in degrees.

(a) (i) $\frac{\pi}{3}$ (ii) $\frac{\pi}{4}$

(b) (i) $\frac{5\pi}{6}$ (ii) $\frac{2\pi}{3}$

(c) (i) $\frac{3\pi}{2}$ (ii) $\frac{5\pi}{3}$

(d) (i) 1.22 (ii) 4.63

6. (a) (i) Express 42° in radians to 3 decimal places.

(ii) Express 168° in radians correct to 3 decimal places.

(b) (i) Express 202.5° in radians in terms of π.

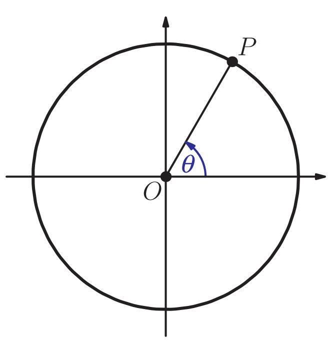
(ii) Express 67.5° in radians in terms of π.

(c) (i) Express $\frac{5\pi}{12}$ radians in degrees.

(ii) Express $\frac{11\pi}{20}$ radians in degrees.

(d) (i) Express 1.62 radians in degrees to one decimal place.

(ii) Express 2.73 radians in degrees to one decimal place.



7. The diagram shows point P on the unit circle corresponding to angle θ (measured in degrees).

Copy the diagram and mark the points corresponding to the following angles.

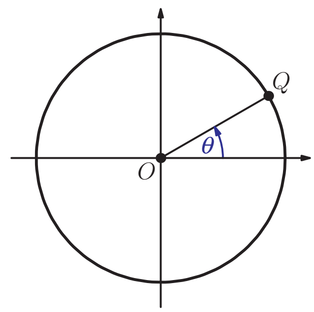
(a) (i) $180° - \theta$ (ii) $180° + \theta$

(b) (i) $\theta + 180°$ (ii) $\theta + 90°$

(c) (i) $90° - \theta$ (ii) $270° - \theta$

(d) (i) $\theta - 360°$ (ii) $\theta + 360°$

8. The diagram shows point Q on the unit circle corresponding to angle θ (measured in radians).



Copy the diagram and mark the points corresponding to the following angles.

(a) (i) $2\pi - \theta$ (ii) $\pi - \theta$

(b) (i) $\theta + \pi$ (ii) $-\pi - \theta$

(c) (i) $\frac{\pi}{2} + \theta$ (ii) $\frac{\pi}{2} - \theta$

(d) (i) $\theta - 2\pi$ (ii) $\theta + 2\pi$

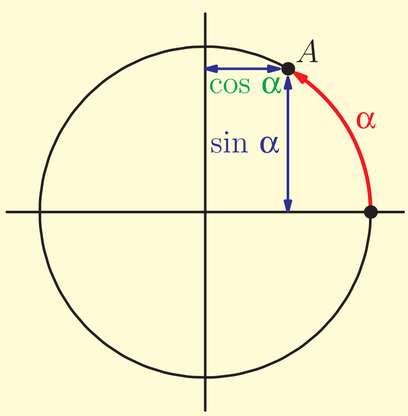
9B Definitions and graphs of sine and cosine functions

We will now define trigonometric functions. Starting with a real number α, first mark a point A on the unit circle representing the number α. The sine and cosine of this number is defined in terms of distance to the axes.

KEY POINT 9.2

The **sine** of the number α, written $\sin \alpha$, is the distance of the point A from the horizontal axis.

The **cosine** of the number α, written $\cos \alpha$, is the distance of the point A from the vertical axis.



You have previously seen sine and cosine defined using right-angled triangles.

See Prior learning Section U on the CD-ROM

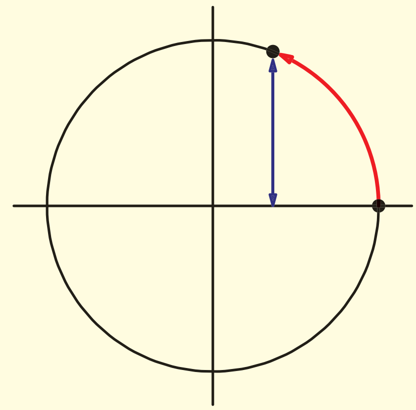
The definition given here is consistent with this but it allows us to define sine and cosine for values beyond 90°. This leads to the question about what makes something a definition. Is it the one that came first historically, the one that is more understandable or the one that is easier to generalise?

Worked example 9.3

By marking the corresponding points on the unit circle, estimate the value of:

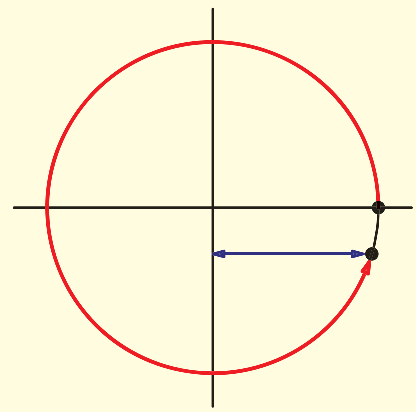
(a) $\sin 1.2$ (b) $\cos 6$ (c) $\cos 2.4$

(a) $\frac{\pi}{2} \approx 1.6$, so the point is in the first quadrant, close to the top



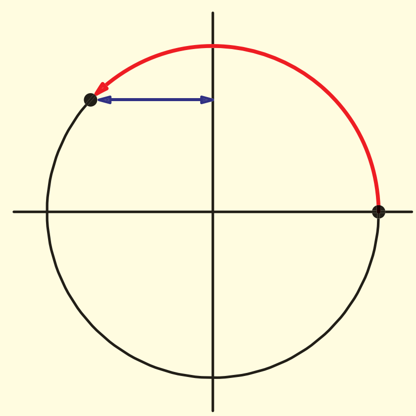
$\sin 1.2 \approx 0.9$

(b) $2\pi \approx 6.2$, so the point is just below the horizontal axis



$\cos 6 \approx 0.95$

(c) $\frac{\pi}{2} \approx 1.6$ and $\pi \approx 3.1$ so the point is around the middle of the second quadrant



$\cos 2.4 \approx -0.7$

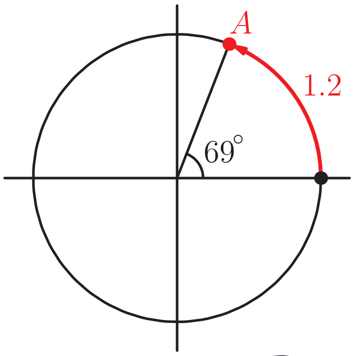
Notice that in the last example $\cos 2.4$ was a negative number. This is because the point corresponding to number 2.4 is to the left of the vertical axis, so we take the distance to be negative.

You may be wondering why we had to change the calculator into radian mode when we are working with real numbers rather than angles. Remember how we defined the radian measure: the size of the angle in radians is equal to the distance travelled by a point around the unit circle as it rotates through that angle. So measuring an angle in radians is the same as measuring the distance 'wrapped' around the circle, which corresponds to real numbers.

EXAM HINT

You can find the values of sine and cosine functions using your calculator. Make sure your calculator is in radian mode.

However, we can also think of a point on the unit circle as corresponding to an angle measured in degrees. For example, point A on the diagram corresponds to the real number 1.2, to an angle of 1.2 radians and to an angle of 69°. If you change your calculator into degree mode and calculate $\sin 69^\circ$, you will get the same answer as when you worked out $\sin 1.2$ in radian mode. Although in applications the sine and cosine functions are frequently used in situations with angles, you should be aware that they are functions which can operate on any real number.



We can use the symmetry of the circle to see how the values of sine and cosine functions of different numbers are related to each other.

EXAM HINT

You should always use radians unless explicitly told to use degrees.

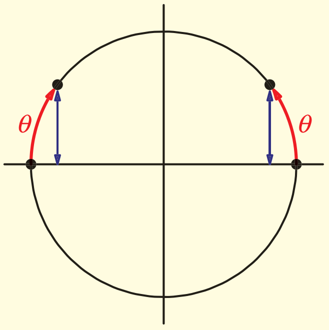
Worked example 9.4

Given that $\sin\theta = 0.6$, find the values of:

(a) $\sin(\pi - \theta)$ (b) $\sin(\theta + \pi)$

Mark the points corresponding to θ and $\pi - \theta$ on the circle

(a)



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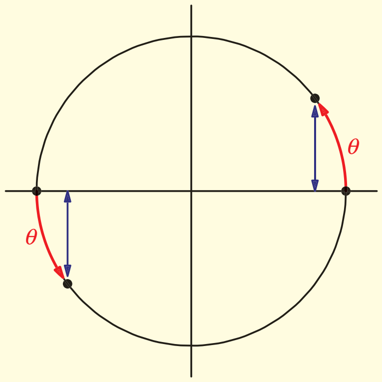
Compare the distances from the horizontal axis

The points are the same distance from the horizontal axis, so $\sin(\pi - \theta) = 0.6$

Mark the points corresponding to θ and $\pi + \theta$ on the circle

(b)

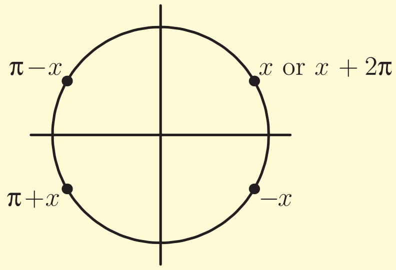


Compare the distances from the horizontal axis

The points are the same distance from the horizontal axis, but the second one is below the axis, so $\sin(\theta + \pi) = -0.6$

The above example illustrates some of the properties of the sine function. Similar properties can be derived for the cosine function. It may be useful to remember those results, summarised below. They can all be derived using circle diagrams.

KEY POINT 9.3



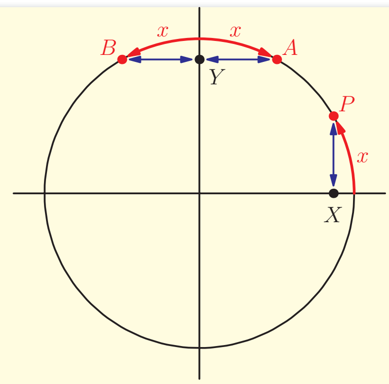
$$\sin x = \sin(\pi - x) = \sin(x + 2\pi)$$
$$\sin(\pi + x) = \sin(-x) = -\sin x$$
$$\cos x = \cos(-x) = \cos(x + 2\pi)$$
$$\cos(\pi - x) = \cos(\pi + x) = -\cos x$$

Worked example 9.5

Given that $\sin x = 0.4$, find the value of:

(a) $\cos\left(\frac{\pi}{2} - x\right)$ (b) $\cos\left(x + \frac{\pi}{2}\right)$

Label the points corresponding to x, $\frac{\pi}{2} - x$ and $x + \frac{\pi}{2}$ on the circle



$\left(\frac{\pi}{2} - x\right)$ is represented by point A, and $AY = PX$

(a) $\cos\left(\frac{\pi}{2} - x\right) = 0.4$

$\left(x + \frac{\pi}{2}\right)$ is represented by point B, and $BY = PX$, but B is to the left of the vertical axis

(b) $\cos\left(x + \frac{\pi}{2}\right) = -0.4$

The above example illustrates a relationship between sine and cosine functions. It may be a good idea to remember, or know how to derive, the following results:

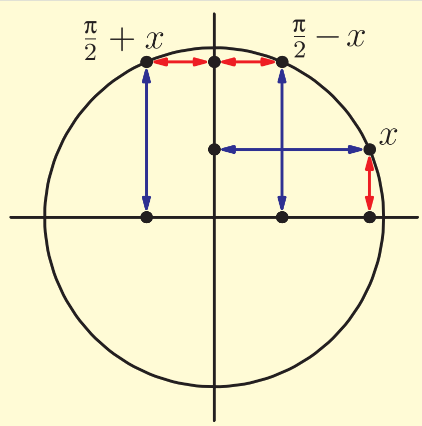
KEY POINT 9.4

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\cos\left(\frac{\pi}{2} + x\right) = -\sin x$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$\sin\left(\frac{\pi}{2} + x\right) = \cos x$$



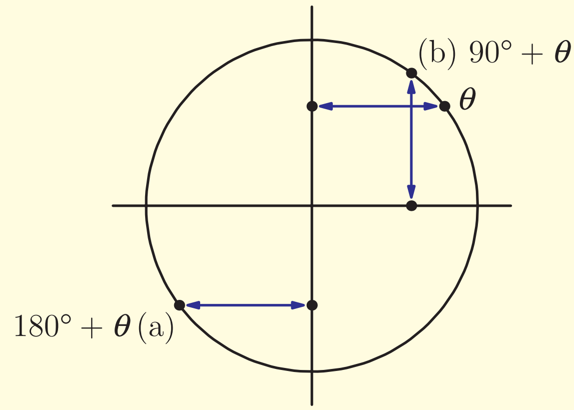
Similar results can be derived when θ represents an angle measured in degrees.

Worked example 9.6

Given that θ is an angle measured in degrees such that $\cos\theta = 0.8$ find the value of

(a) $\cos(180° + \theta)$ (b) $\sin(90° - \theta)$

Mark the points corresponding to angles θ, $180° + \theta$ and $90° - \theta$ on the circle



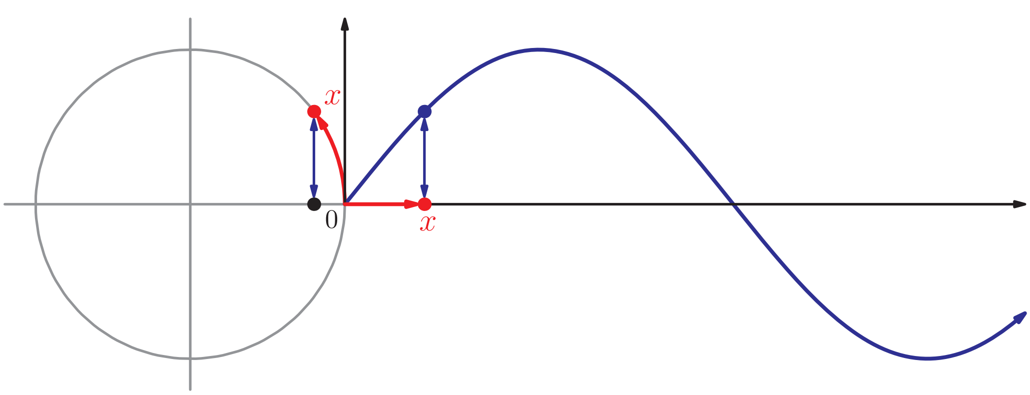
The distance from the vertical axis is 0.8, but the point is to the left of the axis

(a) $\cos(180° + \theta) = -0.8$

By reflection symmetry in the diagonal line, the distance from the *horizontal* axis is 0.8

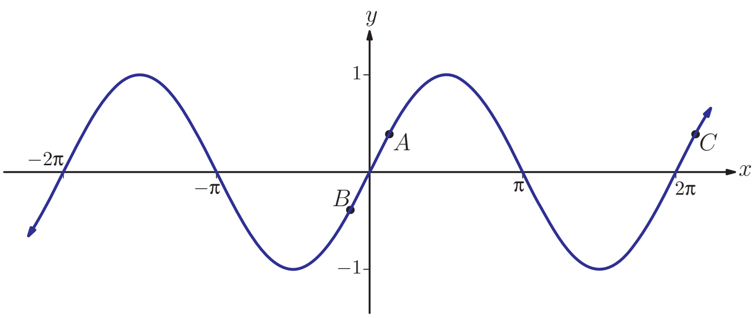
(b) $\sin(90° + \theta) = 0.8$

We have now defined the sine function for all real numbers, so we can draw its graph. To do this, we go back to thinking about the real number line wrapped around the unit circle. Each real number corresponds to a point on the circle, and the value of the sine function is the distance of the point from the horizontal axis.



All of the properties of the sine function discussed above can be seen from the graph. For example, increasing x by 2π corresponds to making a full turn around the circle and returning to the same point. Therefore $\sin(x+2\pi)=\sin x$. We say that the sine function is **periodic** with **period** 2π. By considering points A and B on the graph below, we can see that $\sin(-x)=-\sin x$. We can also see that the minimum possible value of $\sin x$ is -1 and the maximum value is 1. We say that the sine function has **amplitude** 1.

You may wonder how your calculator finds the values of sine and cosine functions. This is explored in Supplementary sheet 8 'How does your calculator work out sin and cos?' on the CD-ROM.

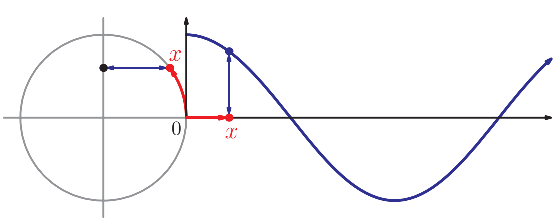


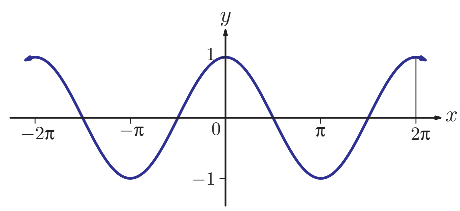
KEY POINT 9.5

A function is periodic if it repeats regularly. The distance between repeats is called the period.

The amplitude is half of the distance between the maximum and minimum value of a function.

To draw the graph of the cosine function we look at the distance of the point representing real number x from the vertical axis.





We can again see many of the properties discussed above on this graph; for example, $\cos(-x)=\cos x$ and $\cos(\pi-x)=-\cos x$. The period and the amplitude are the same as for the sine function. In fact, the graphs of the sine and the cosine functions are related to each other: the graph of $y=\cos x$ is obtained from the graph of $y=\sin x$ by translating it $\frac{\pi}{2}$ units to the left. This again corresponds to one of the properties listed above: $\cos x=\sin\left(x+\frac{\pi}{2}\right)$.

You can see from the graphs that $\sin x$ is an odd function and $\cos x$ is an even function. Both graphs have translational symmetry. We discussed these symmetries in Section 6G.

Trigonometric functions can be used to define **polar coordinates**. This alternative to the Cartesian coordinate system makes it easier to write equations of certain graphs. Such equations produce some beautiful curves, such as the cardioid and the polar rose.

The key properties of sine and cosine functions are summarised below.

KEY POINT 9.6

The sine and cosine functions are periodic with period 2π:

$$\sin x = \sin(x \pm 2\pi) = \sin(x \pm 4\pi) = \ldots$$

$$\cos x = \cos(x \pm 2\pi) = \cos(x \pm 4\pi) = \ldots$$

The sine and cosine functions have amplitude 1:

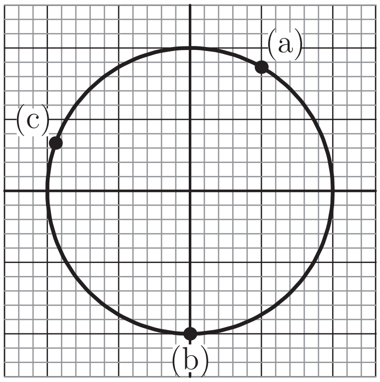
$$-1 \le \sin x \le 1 \text{ and } -1 \le \cos x \le 1$$

The cosine graph is obtained by translating the sine graph $\frac{\pi}{2}$ units to the left:

$$\cos x = \sin\left(x + \frac{\pi}{2}\right) \text{ and } \sin x = \cos\left(x - \frac{\pi}{2}\right)$$

Exercise 9B

1. Write down the approximate values of $\sin x$ and $\cos x$ for the number x corresponding to each of the points marked on the diagram



2. Use the unit circle to find the value of:

(a) (i) $\sin\frac{\pi}{2}$ (ii) $\sin 2\pi$

(b) (i) $\cos 0$ (ii) $\cos(-\pi)$

(c) (i) $\sin\left(-\frac{\pi}{2}\right)$ (ii) $\cos\frac{5\pi}{2}$

3. Use the unit circle to find the value of:

(a) (i) $\cos 90°$ (ii) $\cos 180°$

(b) (i) $\sin 270°$ (ii) $\sin 90°$

(c) (i) $\sin 270°$ (ii) $\cos 450°$

4. Given that $\cos\frac{\pi}{5} = 0.809$ find the value of:

(a) $\cos\frac{4\pi}{5}$ (b) $\cos\frac{21\pi}{5}$

(c) $\cos\frac{9\pi}{5}$ (d) $\cos\frac{6\pi}{5}$

5. Given that $\sin\frac{2\pi}{3} = 0.866$ find the value of:

(a) $\sin\left(\frac{-2\pi}{3}\right)$ (b) $\sin\frac{4\pi}{3}$

(c) $\sin\frac{10\pi}{3}$ (d) $\sin\frac{\pi}{3}$

6. Given that $\cos 40° = 0.766$ find the value of:

(a) $\cos 400°$ (b) $\cos 320°$

(c) $\cos(-220°)$ (d) $\cos 140°$

7. Given that $\sin 130° = 0.766$ find the value of:

(a) $\sin 490°$ (b) $\sin 50°$

(c) $\sin(-130°)$ (d) $\sin 230°$

8. Sketch the graph of $y = \sin x$ for:

(a) (i) $0° \le x \le 180°$ (ii) $90° \le x \le 360°$

(b) (i) $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ (ii) $-\pi \le x \le 2\pi$

9. Sketch the graph of $y = \cos x$ for:

(a) (i) $-180° \le x \le 180°$ (ii) $0° \le x \le 270°$

(b) (i) $\frac{\pi}{2} \le x \le \frac{3\pi}{2}$ (ii) $-\pi \le x \le 2\pi$

10. (a) On the unit circle, mark the points representing $\frac{\pi}{6}, \frac{\pi}{3}$ and $\frac{2\pi}{3}$.

(b) Given that $\sin\frac{\pi}{6} = 0.5$, find the value of:

(i) $\cos\frac{\pi}{3}$ (ii) $\cos\frac{2\pi}{3}$

11. Use your calculator to evaluate the following, giving your answers to 3 significant figures.

(a) (i) $\cos 1.25$ (ii) $\sin 0.68$

(b) (i) $\cos(-0.72)$ (ii) $\sin(-2.35)$

12. Use your calculator to evaluate the following, giving your answers to 3 significant figures.

(a) (i) $\sin 42°$ (ii) $\cos 168°$

(b) (i) $\sin(-50°)$ (ii) $\cos(-227°)$ *[4 marks]*

13. Evaluate $\cos(\pi + x) + \cos(\pi - x)$.

14. Simplify the following expression:

$$\sin x + \sin\left(x + \frac{\pi}{2}\right) + \sin(x + \pi) + \sin\left(x + \frac{3\pi}{2}\right) + \sin(x + 2\pi)$$

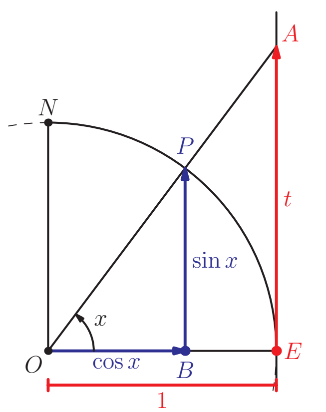
[5 marks]

9C Definition and graph of the tangent function

We now define another trigonometric function – the **tangent** function. It is defined as the ratio between the sine and the cosine functions.

KEY POINT 9.7

$$\tan x = \frac{\sin x}{\cos x}$$

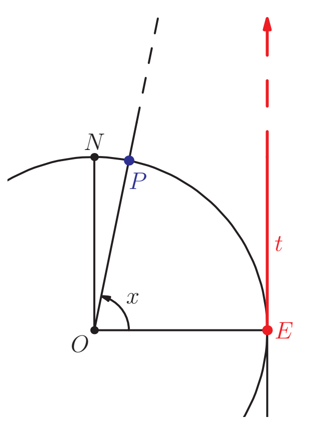


You may wonder why the function is called the tangent function. Consider the diagram alongside:

A tangent is drawn from the point E, where the unit circle meets the positive x-axis. The line at angle x is continued until it hits the tangent. Since *OPB* and *OAE* are similar triangles, we get that:

$$\frac{t}{1} = \frac{\sin x}{\cos x}$$

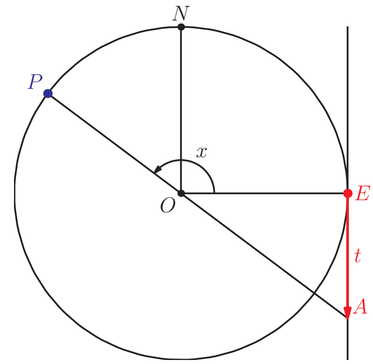
$$\therefore t = \tan x$$



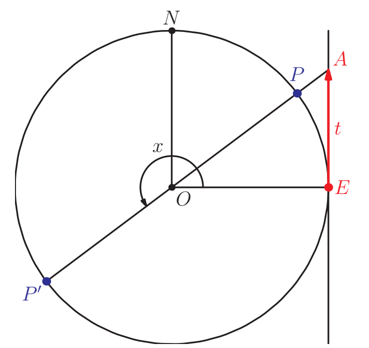
This can be used to see several important properties of the tangent function.

As P moves from E to N, the distance along the tangent increases. When P is close to N, the tangent gets extremely large. If P is actually at N the line ON is parallel to the tangent at E so there is no intersection, and $\tan x$ is undefined. This corresponds in the definition to the situation when $\cos x$ is zero, which is not allowed. In fact, the tangent function is undefined for $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \ldots$ These lines are vertical asymptotes of the graph of $y = \tan x$.

When P moves beyond N the intersection is now below the x-axis and the value of $\tan x$ is negative.



If P' is on the opposite side of the circle to P, it still intersects the tangent at the same point. This means that $\tan x$ repeats every π radians or $180°$.

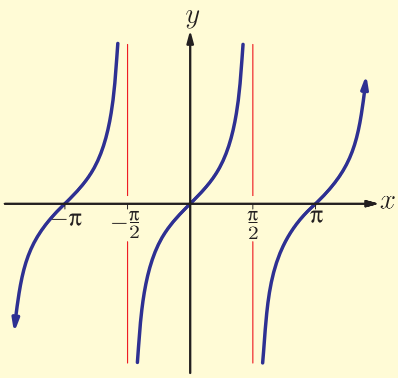


It is equal to zero when $x = 0, \pi, 2\pi \ldots$

This information can be used to sketch the graph of $y = \tan x$.

KEY POINT 9.8

The graph of the tangent function:



Asymptotes are discussed in Section 4C.

We should remember that the points on the unit circle can also represent angles measured in degrees.

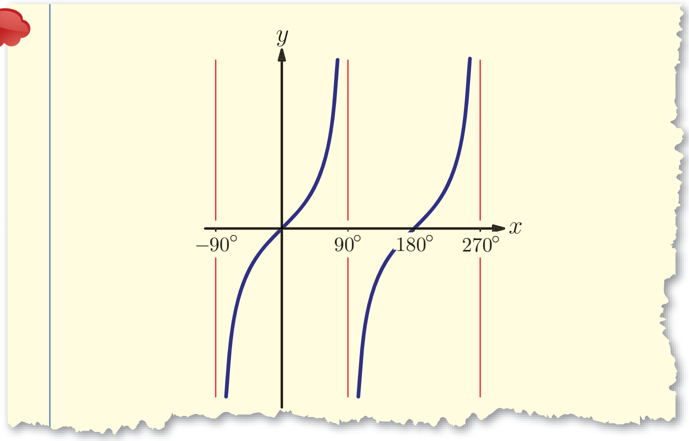
Worked example 9.7

Sketch the graph of $y = \tan x$ for $-90° < x < 270°$.

Find the values for which $\tan x$ is not defined	$\tan x$ undefined: $\cos x = 0$ when $x = -90°, 90°, 270°$.
When is the function positive / negative? When is it zero?	$\tan x$ is: negative in the fourth quadrant, positive in the first quadrant, negative in the second quadrant, ... $\tan x = 0$: $x = 0°$ and $180°$

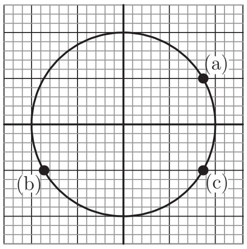
continued . . .

Start by marking the asymptotes and zeros



Exercise 9C

1. By estimating the values of sin θ and cos θ, find the approximate value of tan θ for the points shown on the diagram.



2. Sketch the graph of $y = \tan x$ for:

   (a) (i) $0° \le x \le 360°$ (ii) $-90° \le x \le 270°$

   (b) (i) $\dfrac{\pi}{2} \le x \le \dfrac{5\pi}{2}$ (ii) $-\pi \le x \le \pi$

3. Use your calculator to evaluate the following, giving your answers to 2 decimal places.

   (a) (i) tan 1.2 (ii) tan 4.7

   (b) (i) tan (−0.65) (ii) tan (−7.3)

4. Use your calculator to evaluate the following, giving your answers to 3 significant figures.

   (a) (i) tan 32° (ii) tan168°

   (b) (i) tan (−540°) (ii) tan (−128°)

5. Use the properties of sine and cosine to express the following in terms of tan x.

(a) $\tan(\pi - x)$ (b) $\tan\left(x + \frac{\pi}{2}\right)$

(c) $\tan(x + \pi)$ (d) $\tan(x + 3\pi)$

6. Use the properties of sine and cosine to express the following in terms of $\tan\theta°$.

(a) $\tan(-\theta°)$ (b) $\tan(-360° - \theta°)$

(c) $\tan(-90° - \theta°)$ (d) $\tan(-180° - \theta°)$

7. Sketch the graph of:

(a) $y = 2\sin x - \tan x$ for $-\frac{\pi}{2} \le x \le 2\pi$

(b) $y = 3\cos x + \tan x$ for $-\pi \le x \le \pi$

8. Find the zeros of the following functions:

(a) $y = 2\tan x° + \sin x°$ for $0 \le x \le 360$

(b) $y = 3\cos x° - \tan x°$ for $-180 \le x \le 360$

9. Find the coordinates of maximum and minimum points on the following graphs:

(a) $y = 3\sin x - \tan x$ for $0 \le x \le 2\pi$

(b) $y = \cos x - 2\sin x$ for $-\pi \le x \le \pi$

10. Find approximate solutions of the following equations, giving your answers correct to 3 significant figures.

(a) $\cos x - \tan x = 3,\ x \in]0, 2\pi[$

(b) $\sin x + \cos x = 1,\ x \in [0, 2\pi]$

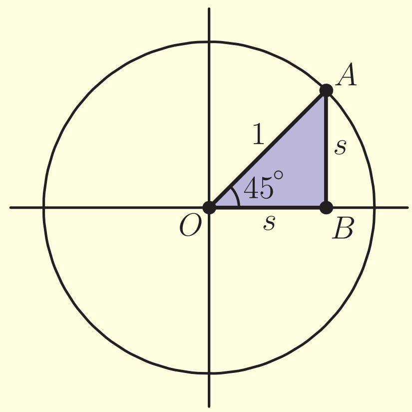
9D Exact values of trigonometric functions

Although values of trigonometric functions are generally difficult to find without a calculator, there are a few numbers for which exact values can be found. The method relies on properties of some special right-angled triangles.

Worked example 9.8

Find the exact values of $\sin\frac{\pi}{4}$, $\cos\frac{\pi}{4}$ and $\tan\frac{\pi}{4}$.

Mark the point corresponding to $\frac{\pi}{4}$ on the unit circle



Look at the triangle *OAB*. It is a right angle triangle, with the angle at *O* equal to 45° (because $\frac{\pi}{4}$ is one eighth of a full turn)

$$s^2 + s^2 = 1, \text{ so } s = \frac{1}{\sqrt{2}}$$

$$\therefore \sin\frac{\pi}{4} = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

We can now use the definition of tan *x*

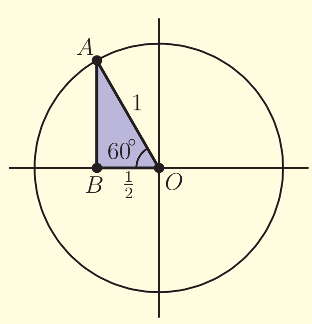
$$\tan\frac{\pi}{4} = \frac{\sin\frac{\pi}{4}}{\cos\frac{\pi}{4}} = 1$$

The other special right-angled triangle is made by cutting an equilateral triangle in half.

Worked example 9.9

Find the exact values of the three trigonometric functions of $\frac{2\pi}{3}$.

Mark the point corresponding to $\frac{2\pi}{3}$ on the unit circle. This is one third of a full turn, so angle *AOB* is 60°



continued . . .

Triangle AOB is half of an equilateral triangle with side 1. OB is equal to half the side of the triangle

$OB = \frac{1}{2}$, point A is to the left of the vertical axis, so

$\cos\frac{2\pi}{3} < 0$

$\therefore \cos\frac{2\pi}{3} = -\frac{1}{2}$

To find AB, use Pythagoras

$AB^2 = 1^2 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}\ \therefore \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

Use the definition of tan x

$\tan\frac{2\pi}{3} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$

The results for these and other special values are summarised below. You may be able to remember them all, but you should understand how they are derived, as shown in the above examples.

EXAM HINT

The table is easier to remember if you notice the pattern in the values of both sin and cos: $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$.

KEY POINT 9.9

Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
Degrees	0	30	45	60	90	120	135	150	180
sin x	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
cos x	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1
tan x	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

10 Trigonometric equations and identities

In this chapter you will learn:

- how to solve equations involving trigonometric functions
- about relationships between different trigonometric functions, called identities
- how to use identities to solve more complicated equations.

Introductory problem

Without a calculator solve the equation $2\cos x = 3\tan x$ for $0 \le x < 2\pi$.

When using trigonometric functions to model real life situations we often need to solve equations where the unknown is in the argument of a trigonometric function, for example, $12 - 4\sin\left(\frac{x}{\pi}\right) = 11$, or $\sin^2 x + \cos x = 0.5$. Because trigonometric functions are periodic, such equations can have more than one solution. In this chapter we will see how to find all solutions in a given interval. We will also see that we can use trigonometric identities, which tell us how different trigonometric functions are related to each other, to transform more complicated equations into simpler ones.

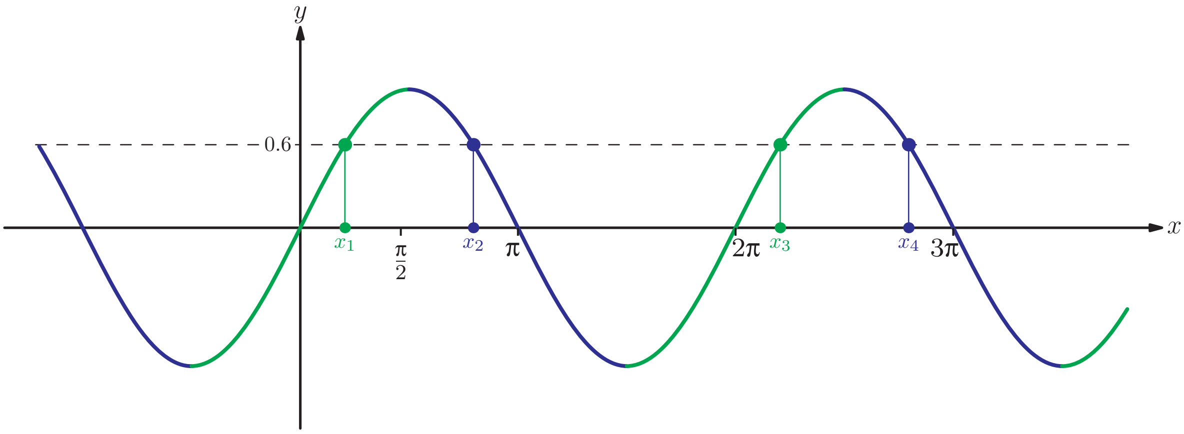
10A Introducing trigonometric equations

We start with the simplest type of trigonometric equation: Suppose we would like to find the x values which satisfy $\sin x = 0.6$.

By applying arcsine to both sides of the equation we get

$x = \arcsin 0.6 = 0.644$ from GDC (3SF).

The arcsine only gives us one solution. However, looking at the graph of $y = \sin x$ we can see that there are many points which satisfy this equation.



The solutions come in pairs – one in the green section of the graph and one in the blue section. The arcsine function will always give us a solution in the green section. To find the solution in the blue section we use fact that the graph has a line of symmetry at $x = \frac{\pi}{2}$. Therefore x_2 is as far below π as x_1 is above zero. In our example, this means that $x_2 = \pi - 0.644 = 2.50$.

Once we have this pair of solutions we can use the fact that the sine graph repeats with period 2π to find the other solutions: $x_3 = x_1 + 2\pi = 6.93$, $x_4 = 2.50 + 2\pi = 8.78$, and so on.

EXAM HINT

When doing subsequent calculations always use the ANS button or the stored value rather than the rounded answer.

See Calculator sheet 1 on the CD-ROM

KEY POINT 10.1

To find the possible values of x satisfying $\sin x = a$:

- use the calculator to find $x_1 = \arcsin a$
- the second solution is given by $x_2 = \pi - x_1$ (or $180 - x_1$ if working in degrees).

Other solutions are found by adding or subtracting 2π (or 360°) to any solution already found.

In the International Baccalaureate® you will only be asked to find solutions in a given interval.

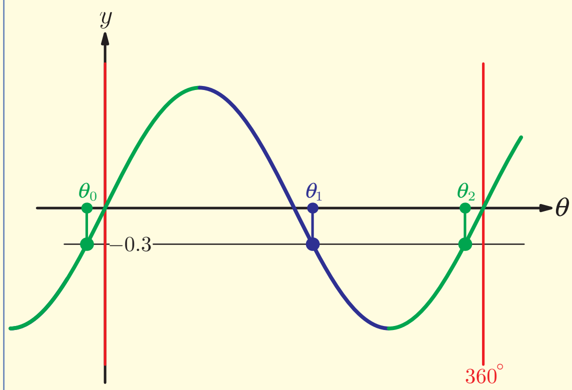
Worked example 10.1

Find the possible values of angle $\theta \in [0, 360°]$ for which $\sin\theta = -0.3$.

Check that the calculator is in degree mode

arcsin (−0.3) = −17.5°

Look at the graph to see how many solutions there are in the required interval



Note how many solutions

There are two solutions.

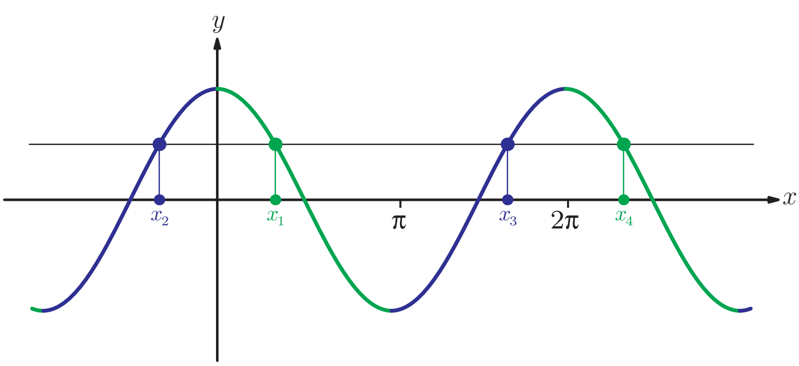
The second solution is given by 180° − θ

$180° - (-17.5°) = 197.5°$

The first solution is not in the required interval, so add 360°

$\theta_1 = 197.5°$
$\theta_2 = 360° + (-17.5°) = 342.5°$

We can apply a similar analysis to the equation $\cos x = a$.



EXAM HINT

Always make sure that your calculator is in the correct mode, degree or radian, as indicated in the question.

The solution x_1 in the green region is given by arccos a. We can use the symmetry of the cosine graph to find x_2; it is the negative of x_1. Once we have this pair of solutions we can use the fact that the cosine graph repeats with period 2π to find the other solutions.

EXAM HINT

It is often useful to know that the two positive solutions will be $\arccos a$ and $2\pi - \arccos a$.

KEY POINT 10.2

To find the possible values of x satisfying $\cos x = a$:

- use the calculator to find $x_1 = \arccos a$
- the second solution is given by $x_2 = -x_1$.

Other solutions are found by adding or subtracting 2π (or 360°) to any solution already found.

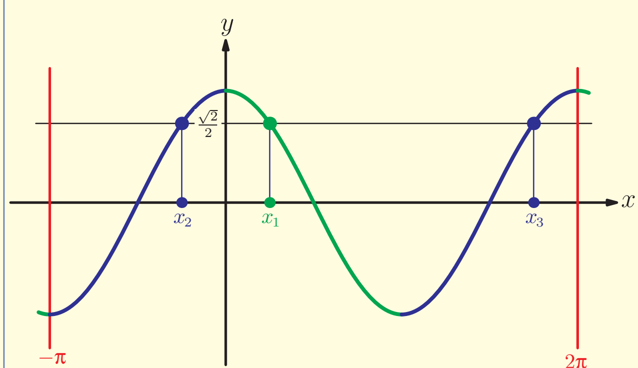
Worked example 10.2

Find the values of x between $-\pi$ and 2π for which $\cos x = \frac{\sqrt{2}}{2}$.

$\arccos \frac{\sqrt{2}}{2}$ is one that we should know

$\arccos\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$

Sketch the graph to see how many solutions there are in the given interval



Note how many solutions

3 solutions

Use the symmetry of the graph to find the other solutions.

$x_1 = \frac{\pi}{4}$

$x_2 = -\frac{\pi}{4}$

$x_3 = 2\pi - \frac{\pi}{4}$

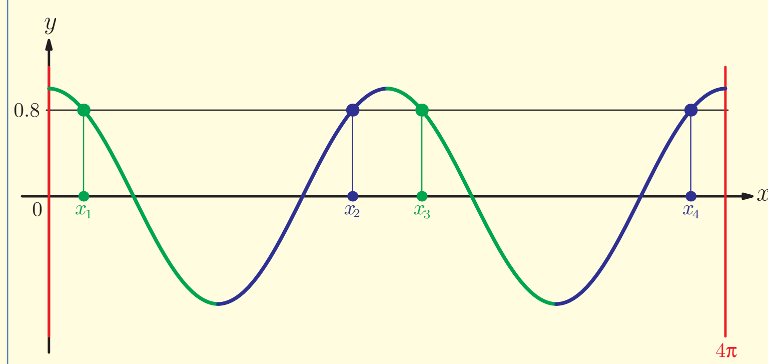
$= \frac{7\pi}{4}$

It can be difficult to know how many times to add or subtract 2π to make sure that we have found all the solutions in a given interval. Drawing a graph first can therefore help, as we can see how many solutions we are looking for and where they are.

Worked example 10.3

Find all the values of x between 0 and 4π for which $\cos x = 0.8$.

Sketch the graph



Note how many solutions

4 solutions

Arccosine on the calculator will give the first value of x

$x_1 = \arccos 0.8 = 0.644$ (3SF)

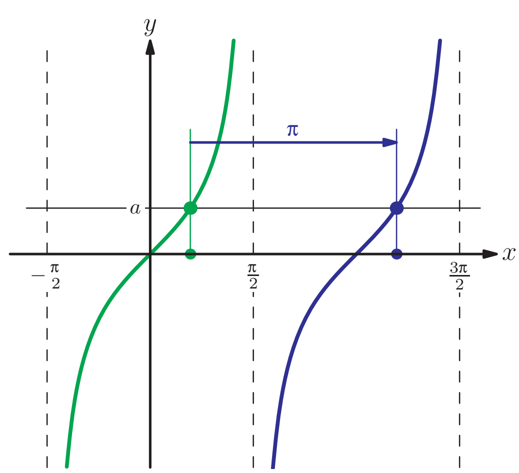
We can use the symmetry of the graph to find the other values

$x_2 = 2\pi - 0.644 = 5.64$ (3SF)

$x_3 = x_1 + 2\pi = 6.93$ (3SF)

$x_4 = x_2 + 2\pi = 11.9$ (3SF)

The procedure for solving equations of the type $\tan x = a$ is slightly different, because the tangent function has period π rather than 2π. It is best seen by looking at the graph of the tangent function.



KEY POINT 10.3

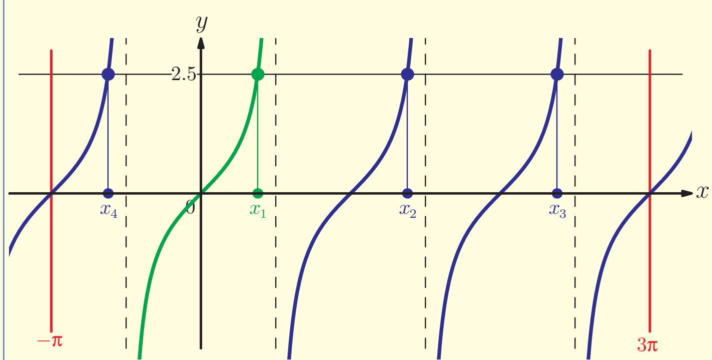
To find the possible values of x satisfying $\tan x = a$:

- use the calculator to find $x_1 = \arctan a$
- other solutions are found by adding or subtracting multiples of π.

Worked example 10.4

Find all real values of $x \in [-\pi, 3\pi]$ such that $\tan x = 2.5$.

Sketch the graph



Note the number of solutions

There are four solutions.

Use a calculator to find the arctan

$x_1 = \arctan 2.5 = 1.19$

The other solutions are found by adding or subtracting π

$x_2 = x_1 + \pi = 4.33$
$x_3 = x_2 + \pi = 7.47$
$x_4 = x_1 - \pi = -1.95$

Exercise 10A

1. Find the exact values of x between 0° and 360° for which:

(a) (i) $\sin x = \frac{1}{2}$ (ii) $\sin x = \frac{\sqrt{2}}{2}$

(b) (i) $\cos x = \frac{1}{2}$ (ii) $\cos x = \frac{\sqrt{3}}{2}$

(c) (i) $\sin x = -\frac{\sqrt{3}}{2}$ (ii) $\sin x = -\frac{1}{2}$

(d) (i) $\tan x = 1$ (ii) $\tan x = \sqrt{3}$

2. Find the exact values of x between 0 and 2π for which:

(a) (i) $\cos x = \frac{\sqrt{3}}{2}$ (ii) $\cos x = \frac{\sqrt{2}}{2}$

(b) (i) $\cos x = -\frac{1}{2}$ (ii) $\cos x = -\frac{\sqrt{3}}{2}$

(c) (i) $\sin x = \frac{\sqrt{2}}{2}$ (ii) $\sin x = \frac{\sqrt{3}}{2}$

(d) (i) $\tan x = \frac{1}{\sqrt{3}}$ (ii) $\tan x = -1$

3. Solve these equations in the given interval, giving your answers to one decimal place.

(a) (i) $\sin x = 0.45$ for $x \in [0°, 360°]$

(ii) $\sin x = 0.7$ for $x \in [0°, 360°]$

(b) (i) $\cos x = -0.75$ for $-180° \le x \le 180°$

(ii) $\cos x = -0.2$ for $-180° \le x \le 180°$

(c) (i) $\tan\theta = \frac{1}{3}$ for $0° \le \theta \le 720°$

(ii) $\tan\theta = \frac{4}{3}$ for $0° \le \theta \le 720°$

(d) (i) $\sin t = -\frac{2}{3}$ for $t \in [-180°, 360°]$

(ii) $\sin t = -\frac{1}{4}$ for $t \in [-180°, 360°]$

4. Solve these equations in the given interval, giving your answers to three significant figures.

(a) (i) $\cos t = \frac{4}{5}$ for $t \in [0, 4\pi]$

(ii) $\cos t = \frac{2}{3}$ for $t \in [0, 4\pi]$

(b) (i) $\sin\theta = -0.8$ for $\theta \in [-2\pi, 2\pi]$

(i) $\sin\theta = -0.35$ for $\theta \in [-2\pi, 2\pi]$

(c) (i) $\tan\theta = -\frac{2}{3}$ for $-\pi \le \theta \le \pi$

(ii) $\tan\theta = -3$ for $-\pi \le \theta \le \pi$

(d) (i) $\cos\theta = 1$ for $\theta \in [0, 4\pi]$

(ii) $\cos\theta = 0$ for $\theta \in [0, 4\pi]$

5. Solve the following equations in the given interval, giving exact answers.

(a) (i) $\sin x = \frac{1}{2}$ for $0° \le x \le 360°$

(ii) $\sin x = \frac{\sqrt{2}}{2}$ for $0° \le x \le 360°$

(b) (i) $\cos x = -1$ for $-180° \le x \le 180°$

(ii) $\sin x = -1$ for $-180° \le x \le 180°$

(c) (i) $\tan x = \sqrt{3}$ for $0° < x < 360°$

(ii) $\tan x = 1$ for $0° < x < 360°$

(d) (i) $\cos x = -\frac{\sqrt{2}}{2}$ for $-360° \le x \le 360°$

(ii) $\cos x = -\frac{\sqrt{3}}{2}$ for $-360° \le x \le 360°$

6. Find the exact solutions of the following equations.

(a) (i) $\cos\theta = \frac{1}{2}$ for $-2\pi < \theta < 2\pi$

(ii) $\cos\theta = \frac{\sqrt{3}}{2}$ for $-2\pi < \theta < 2\pi$

(b) (i) $\sin\theta = -\frac{\sqrt{3}}{2}$ for $-\pi < \theta < 3\pi$

(ii) $\sin\theta = -\frac{\sqrt{2}}{2}$ for $-\pi < \theta < 3\pi$

(c) (i) $\tan\theta = -\frac{1}{\sqrt{3}}$ for $-\pi < \theta < \pi$

(ii) $\tan\theta = -1$ for $-\pi < \theta < \pi$

(d) (i) $\cos\theta = 0$ for $0 < \theta < 3\pi$

(ii) $\sin\theta = 0$ for $0 < \theta < 3\pi$

(e) $\sin\theta = \frac{\sqrt{2}}{2}$ for $-2\pi < \theta < 0$

7. Solve the following equations. Do not use the graphs on your calculator.

(a) (i) $2\sin\theta + 1 = 1.2$ for $0° < \theta < 360°$

(ii) $4\sin x + 3 = 2$ for $-90° < x < 270°$

(b) (i) $3\cos x - 1 = \frac{1}{3}$ for $0 < x < 2\pi$

(ii) $3\cos x - 1 = \frac{1}{3}$ for $0 < x < 2\pi$

(c) (i) $3\tan t - 1 = 4$ for $-\pi < t < \pi$

(ii) $5\tan t - 3 = 8$ for $0 < t < 2\pi$

8. Find the exact values of $x \in (-\pi, \pi)$ for which $2\sin x + 1 = 0$. *[5 marks]*

10B Harder trigonometric equations

In this section we shall look at two different problems which occur when solving trigonometric equations: equations which need to be rearranged first and equations in which the argument of the trigonometric function is more complicated.

It is not always obvious how to write an equation in the form trigonometric function = constant. There are three tactics which are often used.

1. Look for disguised quadratics
2. Take everything over to one side and factorise
3. Use trigonometric identities

See chapter 4 for a reminder on disguised quadratics and solving equations by factorising.

Squares of trigonometric functions come up very frequently and are given a special notation:

Using identities to solve trigonometric equations is covered in Section 10C.

KEY POINT 10.4

$(\cos x)^2$ is written $\cos^2 x$
$(\sin x)^2$ is written $\sin^2 x$
$(\tan x)^2$ is written $\tan^2 x$

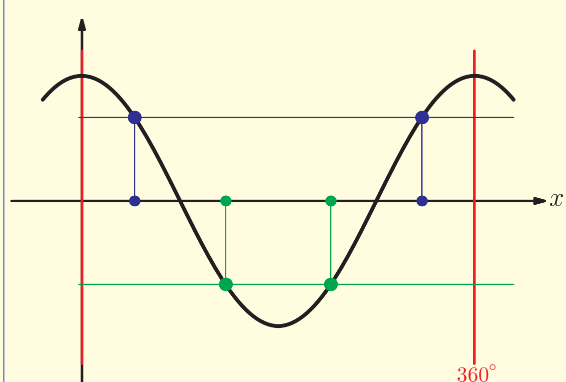
Worked example 10.5

Solve the equation $\cos^2 \theta = \frac{4}{9}$ for $\theta \in [0°, 360°]$. Give answers correct to one decimal place.

First find possible values of $\cos\theta$. Remember ± when we take the square root

$$\cos^2 \theta = \frac{4}{9} \Rightarrow \cos\theta = \pm\frac{2}{3}$$

Draw the graph to see how many solutions there are in the required interval



2 Solutions to each.

continued . . .

Solve each equation separately

When $\cos\theta = \frac{2}{3}$:

$$\arccos\left(\frac{2}{3}\right) = 48.2°$$

$$\theta = 48.2° \text{ or } 360° - 48.2° = 311.8°$$

When $\cos\theta = -\frac{2}{3}$:

$$\arccos\left(-\frac{2}{3}\right) = 131.8°$$

$$\theta = 131.8° \text{ or } 360° - 131.8° = 228.2°$$

List all the solutions.

$\theta_1 = 48.2°$
$\theta_2 = 131.8°$
$\theta_3 = 228.2°$
$\theta_4 = 311.8°$

In the next example we need to use factorising.

Worked example 10.6

Solve the equation $3\sin x\cos x = 2\sin x$ for $-\pi \le x \le \pi$.

We cannot find the inverse sine or inverse cosine directly. However, both sides have a factor of sin *x*, so we can make the equation equal to zero and factorise

$$3\sin x\cos x - 2\sin x = 0$$
$$\sin x(3\cos x - 2) = 0$$

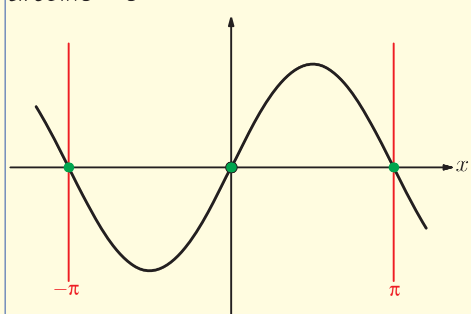
If a product is equal to 0, then one of the factors must be equal to 0

$$\sin x = 0 \text{ or } \cos x = \frac{2}{3}$$

We can now solve each equation separately. Draw the graph for each equation to see how many solutions there are

When $\sin x = 0$:

$\arcsin 0 = 0$

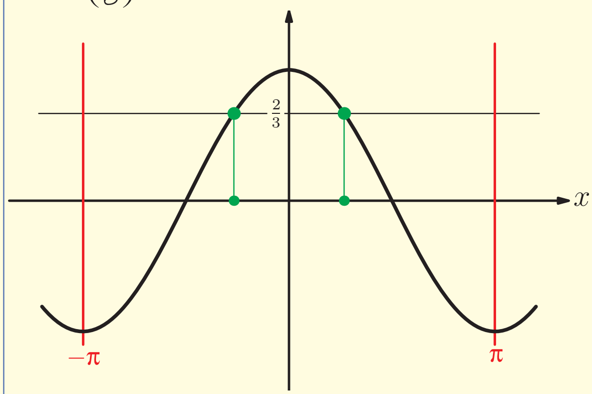


$$x = 0 \text{ or } \pi - 0 = \pi \text{ or } \pi - 2\pi = -\pi$$

continued . . .

When $\cos x = \frac{2}{3}$:

$\arccos\left(\frac{2}{3}\right) = 0.841$



$2\pi - 0.841 = 5.44$ is not in the interval

$5.44 - 2\pi = -0.841$ is in the interval

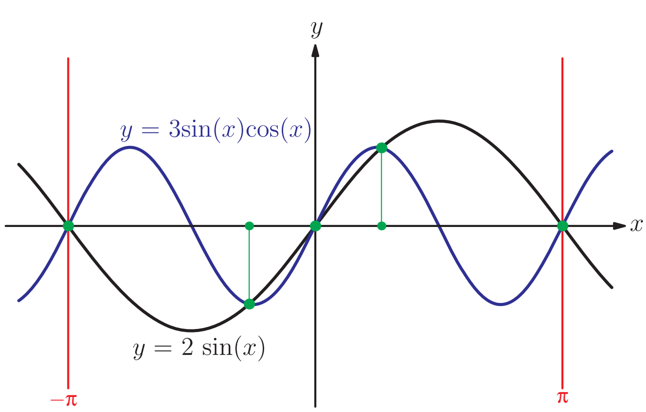
We have found 5 solutions

$x = -\pi,\ -0.841, 0, 0.841,\ \pi$

EXAM HINT

Do not be tempted to divide both sides of the original equation by sin x or you could lose some solutions, because sin x could be zero.

We can check these solutions using the graph on the calculator. Sketching graphs of $y = 3\sin x \cos x$ and $y = 2\sin x$ in the given interval shows that they intersect at five points.



Another common type of trigonometric equation is a disguised quadratic.

Disguised quadratics are often combined with the use of the Pythagorean identity – see Worked example 10.18.

Worked example 10.7

(a) Given that $3\sin^2 x - 5\sin x + 1 = 0$, find the possible values of sin x.

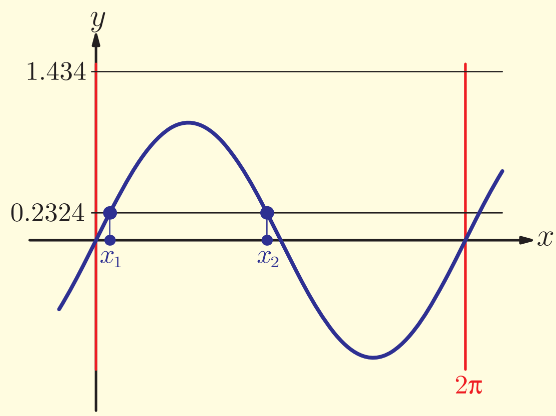
(b) Hence solve the equation $3\sin^2 x - 5\sin x + 1 = 0$ for $0 < x < 2\pi$.

Recognise that this is a quadratic equation in sin x. Since we cannot factorise it, use the quadratic formula

(a) $\sin x = \dfrac{5 \pm \sqrt{5^2 - 4\times3\times1}}{2\times3}$

$\sin x = 1.434$ *or* 0.2324

Draw the graph to see how many solutions there are



There are two solutions.

sin x is always between –1 and 1, so only one of the values is possible

$\sin x = 1.434 > 1$ is impossible.
Hence $\sin x = 0.2324$.

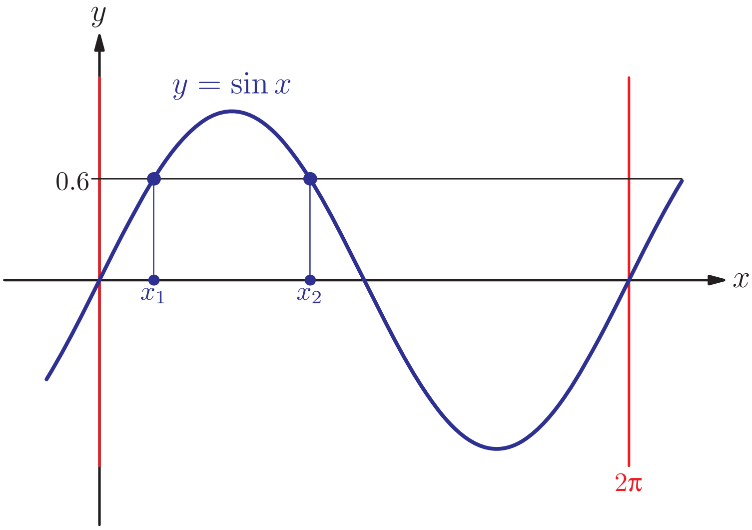
We can now solve the new equation as before

(b) $\arcsin(0.2324) = 0.235$
$x = 0.235$ *or* $\pi - 0.235 = 2.91$

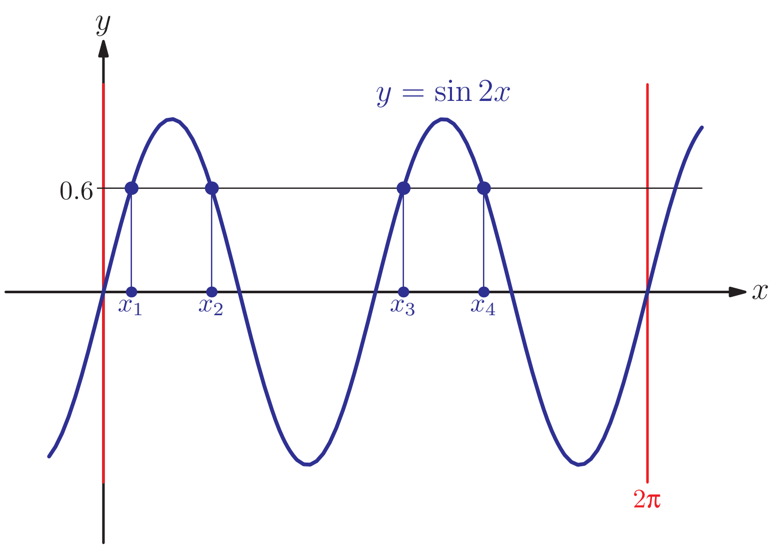
EXAM HINT

Questions like those in Worked example 10.7 force you to use algebraic methods to solve the equation. However, you can still use the graph on your calculator to check your solution. It is particularly useful to make sure that you have not missed out any solutions.

We will now look at equations where the argument of the trigonometric function is more complicated than just x. If we solve the equation $\sin x = 0.6$ for $0 \le x \le 2\pi$ we can see from the graph that there are two solutions:



If we solve the equation $\sin 2x = 0.6$ for $0 \le x \le 2\pi$ we can see from the graph that there are four solutions:



We need to extend the methods from Section 10A to deal with equations like this. A substitution is a useful approach.

Worked example 10.8

Find the zeros of the function $3\sin(2x)+1$ for $x\in[0,2\pi]$.

Write down the equation to be solved

$3\sin(2x)+1=0$

Rearrange it into the form $\sin(A)=k$

$\sin(2x)=-\frac{1}{3}$

Make a substitution for the argument

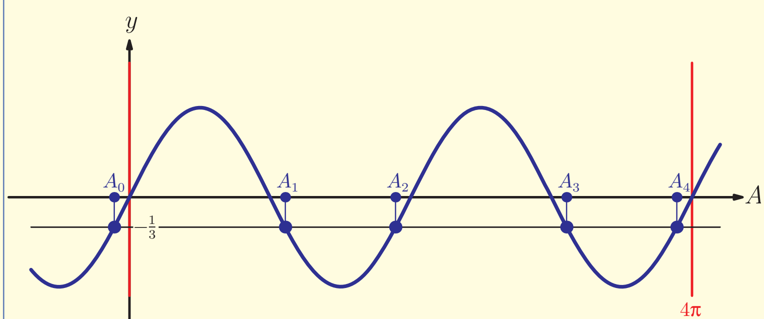
$A=2x$

Rewrite the interval in terms of A

If $x\in[0,2\pi]$ then $A\in[0,4\pi]$.

Solve the equation for A

$\sin A=-\frac{1}{3}$



There are four solutions.

$A_0=\arcsin\left(-\frac{1}{3}\right)=-0.3398$ (outside of interval)

$A_1=\pi-A_0=3.481$
$A_2=A_0+2\pi=5.943$
$A_3=A_1+2\pi=9.764$
$A_4=A_2+2\pi=12.23$

Transform the solutions back into x

$x=\frac{A}{2}$

$=1.74, 2.97, 4.88, 6.11$ (3SF)

This procedure may be summarised in the following four-step process:

KEY POINT 10.5

To solve trigonometric equations:

1. Make a substitution for the **argument** of the trigonometric function (such as $A = 2x$).
2. Change the interval for x into the interval for A.
3. Solve the equation in the usual way.
4. Transform the solutions back into the original variable.

The following example shows this method in a more complicated situation.

Worked example 10.9

Solve the equation $3\cos(2x+1) = 2$ for $x \in [-\pi, \pi]$.

Write the equation in the form $\cos(A) = k$

$$\cos(2x+1) = \frac{2}{3}$$

Make a substitution for the argument

$$A = 2x+1$$

Rewrite the interval

If $-\pi \le x \le \pi$ then

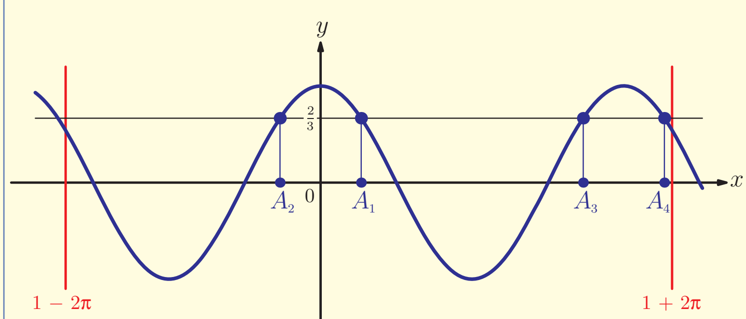
$-2\pi \le 2x \le 2\pi$

$-2\pi + 1 \le 2x+1 \le 2\pi + 1$

So $A \in [-2\pi + 1,\ 2\pi + 1]$

Solve the equation for A

$$\cos A = \frac{2}{3}$$



There are four solutions

$$A_1 = \arccos\left(\frac{2}{3}\right) = 0.841$$

continued . . .

$A_2 = -A_1 = -0.841$
$A_3 = A_2 + 2\pi = 5.44$
$A_4 = A_1 + 2\pi = 7.12$

Transform the solutions back into x

$x = \dfrac{A-1}{2}$

$x = -0.0795, 2.22, 6.2035, -0.921$

We use the final example of this section to revisit the tangent function, working in degrees and finding exact solutions.

Worked example 10.10

Solve the equation $3\tan(\frac{1}{2}\theta - 30°) = \sqrt{3}$ for $0° \leq \theta \leq 720°$.

Rearrange the equation into the form $\tan(A) = k$

$\tan\left(\frac{1}{2}\theta - 30°\right) = \dfrac{\sqrt{3}}{3} = \dfrac{1}{\sqrt{3}}$

Make a substitution for the argument

$A = \frac{1}{2}\theta - 30°$

Rewrite the interval

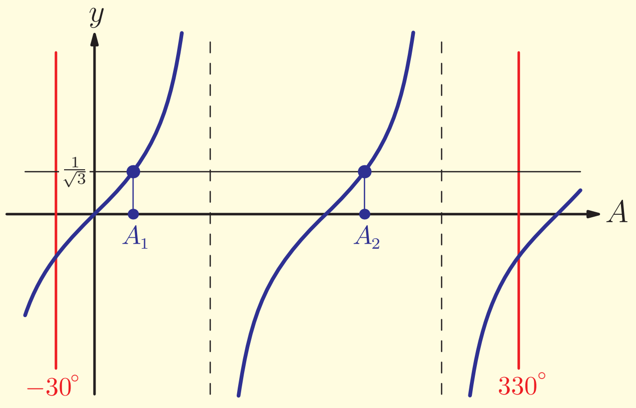
If $0° \leq \theta \leq 720°$ then

$0° \leq \frac{1}{2}\theta \leq 360°$

$-30° \leq \frac{1}{2}\theta - 30° \leq 330°$

Solve the equation for A

$\tan A = \dfrac{1}{\sqrt{3}}$



continued . . .

There are two solutions:

$A_1 = \arctan\left(\frac{1}{\sqrt{3}}\right) = 30°$

$A_2 = A_1 + 180° = 210°$

Transform the solutions back into θ

$\theta = 2(A + 30°)$
$= 120°$ or $480°$

Exercise 10B

1. Solve, giving your answers to 3 significant figures:

   (a) (i) $\tan^2 x = 2$ for $-\pi \le x \le \pi$

   (ii) $\sin^2 x = 0.6$ for $-\pi \le x \le \pi$

   (b) (i) $9\cos^2\theta = 4$ for $0° < \theta < 360°$

   (ii) $3\tan^2\theta = 5$ for $0° < \theta < 360°$

2. Without using a graphing function on your calculator, find all solutions of these equations in the given interval. Use the graphs on your calculator to check your answers.

   (a) (i) $3\sin x - 2\sin x\cos x = 0$ for $0° \le x \le 360°$

   (ii) $4\cos x - \sin x\cos x = 0$ for $0° \le x \le 360°$

   (b) (i) $4\sin^2\theta = 3\sin\theta$ for $\theta \in [-\pi, \pi]$

   (ii) $3\cos^2\theta = -\cos\theta$ for $\theta \in [-\pi, \pi]$

   (c) (i) $\tan^2 t - 5\tan t + 5 = 0$ for $t \in]0, 2\pi[$

   (ii) $2\tan^2 t + \tan t - 1 = 0$ for $t \in]0, 2\pi[$

   (d) (i) $\sin\theta\tan\theta + \frac{1}{2}\tan\theta = 0$ for $\theta \in [0, 2\pi]$

   (ii) $2\cos\theta\tan\theta - 3\cos\theta = 0$ for $\theta \in [0, 2\pi]$

   (e) (i) $2\cos^2 x + 3\cos x = 2$ for $0° < x < 180°$

   (ii) $\cos^2 x - 2\cos x = 3$ for $0° < x < 180°$

3. Solve the following equations in the given interval, giving your answers to 3 significant figures. Do not use the graphing function on your GDC.

   (a) (i) $\cos 2x = \frac{1}{3}$ for $0° \le x \le 360°$

   (ii) $\cos 3x = \frac{2}{5}$ for $0° \le x \le 360°$

(b) (i) $\sin(3x-1) = -0.2$ for $0 \le x \le \pi$

(ii) $\sin(2x+1) = \frac{2}{3}$ for $0 \le x \le 2\pi$

(c) (i) $\tan(x-45°) = 2$ for $-180° \le x \le 180°$

(ii) $\tan(x+60°) = -3$ for $-180° \le x \le 180°$

4. Find the exact solutions in the given interval:

(a) (i) $\sin 2x = \frac{1}{2}$ for $0 \le x \le 2\pi$

(ii) $\sin 3x = -\frac{1}{2}$ for $0 \le x \le 2\pi$

(b) (i) $\cos 2x = -\frac{\sqrt{2}}{2}$ for $0° \le x \le 360°$

(ii) $\cos 3x = \frac{1}{2}$ for $-180° \le x \le 180°$

(c) (i) $\tan 4x = \sqrt{3}$ for $0 \le x \le \pi$

(ii) $\tan 2x = \frac{1}{\sqrt{3}}$ for $0 \le x \le \pi$

5. Find the exact solutions in the given interval:

(a) (i) $\cos(x+60°) = \frac{\sqrt{3}}{2}$ for $0° \le x \le 360°$

(ii) $\cos\left(x-\frac{\pi}{3}\right) = \frac{1}{2}$ for $-\pi \le x \le \pi$

(b) (i) $\sin\left(x-\frac{\pi}{3}\right) = -\frac{1}{2}$ for $-\pi \le x \le \pi$

(ii) $\sin(x-120°) = -\frac{\sqrt{2}}{2}$ for $0° \le x \le 360°$

(c) (i) $\tan\left(x+\frac{\pi}{2}\right) = 1$ for $0 < x < 2\pi$

(ii) $\tan\left(x-\frac{\pi}{4}\right) = -1$ for $0 < x < 2\pi$

6. Solve the equation $3\cos x = \tan x$ for $0 \le x \le 2\pi$. *[8 marks]*

7. (a) Given that $2\sin^2 x - 3\sin x = 2$, find the exact values of $\sin x$.

(b) Hence solve the equation $2\sin^2 x - 3\sin x = 2$ for $0° < x < 360°$. *[6 marks]*

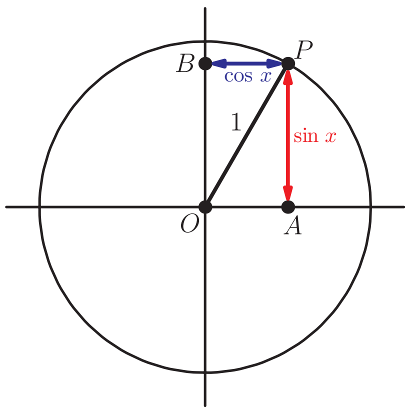
8. Solve the equation $\sin x \tan x = \sin^2 x$ for $-\pi \le x \le \pi$. *[8 marks]*

9. Solve the equation $\sin(x^2) = \frac{1}{2}$ for $-\sqrt{\pi} < x < \sqrt{\pi}$ *[5 marks]*

10C Trigonometric identities

We have already seen one example of an identity in chapter 9: $\frac{\sin x}{\cos x} = \tan x$. The two sides are equal for all values of x (except when $\cos = 0$, where $\tan x$ is undefined). There are many other identities involving trigonometric functions, and we will meet some of them in this section.

Consider again the familiar unit circle diagram, with point P representing number x. According to the definitions of sine and cosine functions, $AP = \sin x$ and $BP = \cos x = OA$.



But triangle OAP is right angled, with hypotenuse 1. Using Pythagoras' Theorem, we get:

KEY POINT 10.6

Pythagorean identity

$$\sin^2 x + \cos^2 x = 1$$

We have already seen examples where we used the values of $\sin x$ and $\cos x$ to find the value of $\tan x$. Using the Pythagorean identity, we only need to know the value of one of the functions to find the other two.

EXAM HINT

Many books write $\frac{\sin x}{\cos x} \equiv \tan x$ to emphasise that the expression is an identity (true for all values of x) rather than an equation (only true for some values of x which need to be found). However, the International Baccalaureate® syllabus and most exam questions use the equals sign in identities, so we will do the same in this book, unless there is a possibility of confusion.

Worked example 10.11

Given that $\sin x = \frac{1}{3}$, find the possible values of $\cos x$ and $\tan x$.

Use identity relating sin and cos

$$\sin^2 x + \cos^2 x = 1$$

$$\left(\frac{1}{3}\right)^2 + \cos^2 x = 1$$

$$\cos^2 x = 1 - \frac{1}{9} = \frac{8}{9}$$

Remember ± when you take square root

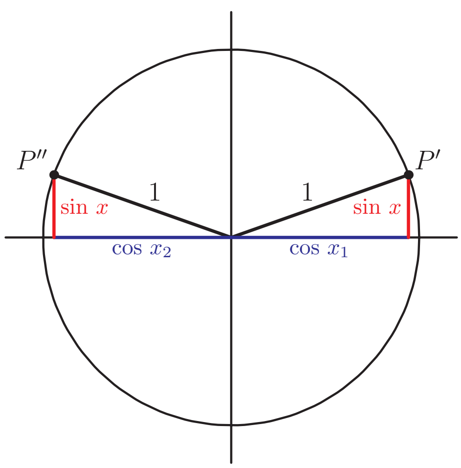
$$\therefore \cos x = \pm\sqrt{\frac{8}{9}} = \pm\frac{2\sqrt{2}}{3}$$

Find tan given sin and cos

$$\tan x = \frac{\sin x}{\cos x}$$

$$\therefore \tan x = \frac{\frac{1}{3}}{\pm\frac{2\sqrt{2}}{3}} = \pm\frac{\sqrt{2}}{4}$$

Notice that for a given value of $\sin x$, there are two possible values of $\cos x$. This can be seen by looking at the unit circle: points P' and P'' are both distance, $\sin x$ above the horizontal axis, but have different values of $\cos x$ (equal in size but opposite in sign).



Also notice that we do not need to know what x is; in fact we know that there are infinitely many possible values for x.

We can specify one of the two possible values by restricting x to be in a particular quadrant.

Worked example 10.12

If $\tan x = 2$ and $\frac{\pi}{2} < x < \pi$ find the value of $\cos x$.

We need to link tan x and cos x. We know $\frac{\sin x}{\cos x} = \tan x$ and can then use $\sin^2 x + \cos^2 x = 1$ to eliminate sin x

$\sin x = \tan x \cos x$

We can substitute sin x from the first identity into the second

$\therefore \tan^2 x \cos^2 x + \cos^2 x = 1$

Now substitute the known value of tan x

$2^2 \cos^2 x + \cos^2 x = 1$

$\Leftrightarrow 5\cos^2 x = 1$

$\Leftrightarrow \cos x = \pm\frac{\sqrt{5}}{5}$

x is in the second quadrant, so cos x is negative

$\cos x < 0$

$\therefore \cos x = -\frac{\sqrt{5}}{5}$

We can use known identities to derive new ones.

For example, multiplying both sides of the Pythagorean identity by 3 gives:

$$3\sin^2 x + 3\cos^2 x = 3$$

We can also rearrange an identity, just as we would with an equation. For example, we can rearrange the Pythagorean identity to get:

$$\cos^2 x = 1 - \sin^2 x \ (1)$$

In chapter 12, we shall meet a more direct way to relate $\tan x$ *and* $\cos x$*.*

We can also rearrange one identity and then substitute into another. For example, we can rearrange $\tan x = \frac{\sin x}{\cos x}$ to $\cos x = \frac{\sin x}{\tan x}$ and substitute it into the identity (1) above to get:

$$\frac{\sin^2 x}{\tan^2 x} = 1 - \sin^2 x$$

Finally, in each of these identities the variable x can be replaced by any other variable or expression, as long as each occurrence of x is replaced by the same thing.

This means that, for example:

$$\sin^2(2x+1)+\cos^2(2x+1)=1$$
$$\text{and } \frac{\sin 2x}{\cos 2x}=\tan 2x$$

By combining these ideas we can derive many more complicated identities, especially if we also use algebraic identities such as the difference of two squares.

Worked example 10.13

Starting from the identity $\sin^2 x+\cos^2 x=1$, derive the identity $\sin^4 x-\cos^4 x=2\sin^2 x-1$.

$\sin^2 x+\cos^2 x=1$

Look for some relationship between the two identities. In the second identity LHS is the difference of two squares:
$a^4-b^4=(a^2-b^2)(a^2+b^2)$

$\sin^4 x-\cos^4 x$

This means that to get the LHS of the second identity, we need to multiply the first identity by $\sin^2 x-\cos^2 x$

$\Rightarrow (\sin^2 x-\cos^2 x)(\sin^2 x+\cos^2 x)=\sin^2 x-\cos^2 x$

$\Rightarrow \sin^4 x-\cos^4 x=\sin^2 x-\cos^2 x$

Remove cos x from RHS, using $\cos^2 x=1-\sin^2 x$

$\Rightarrow \sin^4 x-\cos^4 x=\sin^2 x-(1-\sin^2 x)$
$\Rightarrow \sin^4 x-\cos^4 x=2\sin^2 x-1$

We know that the final equation is true for all x because we started from a known identity and only used known identities and algebraic manipulations. Therefore we have ended up with a new identity.

We have already looked at some methods for proving identities in Section 4H.

The question we want to ask now is, if we are given the final equation, how can we show that it is an identity? One way would be to try and recreate the steps we used to derive it from $\sin^2 x+\cos^2 x=1$. However, this seems to involve a lot of guessing! In the next three examples we look at various techniques for proving identities.

We begin with the same identity we just proved, but without the hint to use $\sin^2 x+\cos^2 x=1$ as a starting point.

Worked example 10.14

Prove the identity $\sin^4 x - \cos^4 x = 2\sin^2 x - 1$.

The task is to show that the two sides are equal. One way to do this is to start from the left-hand side (LHS) and transform it until we get the right-hand side (RHS)

$LHS = \sin^4 x - \cos^4 x$

The LHS involves both sin x and cos x, while the RHS involves only sin x. So it seems like a good idea to replace $\cos^4 x$ on the LHS by using $\cos^2 x = 1 - \sin^2 x$

$$\begin{aligned}&= \sin^4 x - (1 - \sin^2 x)^2 \\ &= \sin^4 x - (1 - 2\sin^2 x + \sin^4 x) \\ &= -1 + 2\sin^2 x \\ &= RHS\end{aligned}$$

The two sides are equal, so we have proved the identity

$LHS = RHS \therefore$ proved

Sometimes it is not obvious how to get from one side to the other. In that case, we can transform each side separately and 'meet in the middle'.

Worked example 10.15

Prove the identity $\dfrac{1}{\cos\theta} - \cos\theta = \sin\theta\tan\theta$.

Write the whole LHS expression as a single fraction

$$LHS = \frac{1}{\cos\theta} - \frac{\cos^2\theta}{\cos\theta} = \frac{1-\cos^2\theta}{\cos\theta}$$

The RHS involves $\sin\theta$, so use $\sin^2\theta + \cos^2\theta$ in the numerator

$$= \frac{\sin^2\theta}{\cos\theta}$$

Look at the RHS. We know an identity for $\tan\theta$

$$RHS = \sin\theta\frac{\sin\theta}{\cos\theta} = \frac{\sin^2\theta}{\cos\theta}$$

The two sides are equal, so we have proved the identity

$LHS = RHS \therefore$ proved

Worked example 10.15 illustrates the importance of being able to simplify expressions with fractions. In a more complicated example like this, it is useful to write what you are doing, so you can check your working more easily.

Worked example 10.16

Prove the identity $\sin x \cos x = \dfrac{\tan x}{1+\tan^2 x}$.

There is no obvious way to simplify the LHS, but the RHS involves $\tan x$, so rewrite in terms of $\sin x$ and $\cos x$

$$RHS = \frac{\frac{\sin x}{\cos x}}{1+\frac{\sin^2 x}{\cos^2 x}} \qquad \left(\text{using } \tan x = \frac{\sin x}{\cos x}\right)$$

To simplify the 'big' fraction, multiply top and bottom by the common denominator, which is $\cos^2 x$

$$= \frac{\frac{\sin x}{\cos x} \times \cos^2 x}{\cos^2 x + \sin^2 x} \qquad (\text{multiplying by } \cos^2 x)$$

$$= \frac{\sin x \cos x}{\cos^2 x + \sin^2 x}$$

Now use $\sin^2 x + \cos^2 x = 1$

$$= \sin x \cos x \qquad (\text{using } \sin^2 x + \cos^2 x = 1)$$

This is equal to the LHS

$$= LHS \quad \therefore \text{proved}$$

Exercise 10C

1. Find the exact values of cos x and tan x given that:

   (a) $\sin x = \frac{1}{3}$ and $0° < x < 90°$

   (b) $\sin x = \frac{4}{5}$ and $0° < x < 90°$

2. Find the exact values of sin θ and tan θ given that:

   (a) $\cos\theta = -\frac{1}{3}$ and $180° < \theta < 270°$

   (b) $\sin\theta = -\frac{3}{4}$ and $180° < \theta < 270°$

3. (a) Find the exact value of cos x if:

   (i) $\sin x = \frac{1}{5}$ and $90° < x < 180°$

   (ii) $\sin x = -\frac{1}{2}$ and $270° < x < 360°$

   (b) Find the exact value of tan x if:

   (i) $\cos x = \frac{3}{5}$ and $-\frac{\pi}{2} < x < 0$

(ii) $\cos x = -1$ and $\frac{\pi}{2} < x < \frac{3\pi}{2}$

4. (a) Find the possible values of $\cos x$ if $\tan x = \frac{2}{3}$.

(b) Find the possible values of $\sin x$ if $\tan x = -\frac{1}{2}$.

5. Find the exact value of:

(a) $3\sin^2 x + 3\cos^2 x$ (b) $\sin^2 5x + \cos^2 5x$

(c) $-2\cos^2 2x - 2\sin^2 2x$ (d) $2\tan^2 2x - \frac{2}{\cos^2 2x}$

(e) $\frac{1}{\sin^2 x} - \frac{1}{\tan^2 x}$ (f) $\frac{3}{2\sin^2 4x} - \frac{3}{2\tan^2 4x}$

6. (a) (i) Express $3\sin^2 x + 4\cos^2 x$ in terms of $\sin x$ only.

(ii) Express $\cos^2 x - \sin^2 x$ in terms of $\cos x$ only.

7. If $t = \tan x$, express the following in terms of t:

(a) $\cos^2 x$ (b) $\sin^2 x$

(c) $\cos^2 x - \sin^2 x$ (d) $\frac{2}{\sin^2 x} + 1$

8. Prove the following identities:

(a) (i) $(\sin x + \cos x)^2 + (\sin x - \cos x)^2 = 2$

(ii) $(2\sin x - \cos x)^2 + (\sin x + 2\cos x)^2 = 5$

(b) (i) $\sin\theta\tan\theta + \cos\theta = \frac{1}{\cos\theta}$

(ii) $\frac{\cos\theta}{\sin\theta} = \frac{1}{\sin\theta} - \sin\theta$

9. Express $3 - 2\tan^2 x$ in terms of $\cos x$ only. *[3 marks]*

10. Express $\frac{1+\tan^2 x}{\cos^2 x}$ in terms of $\sin x$, simplifying your answer. *[4 marks]*

11. Prove the following identities:

(a) $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \frac{1}{\sin\theta\cos\theta}$

(b) $\frac{1}{\cos\theta} + \tan\theta = \frac{\cos\theta}{1-\sin\theta}$ *[8 marks]*

10D Using identities to solve equations

Now we can use trigonometric identities to solve more complicated equations.

Usually we either start by replacing $\tan x$ by $\frac{\sin x}{\cos x}$, or using the Pythagorean identity. The latter can only be used if the equation contains squares and usually results in a quadratic equation.

12A Double angle identities

This section looks at trigonometric functions when the argument of the function doubles.

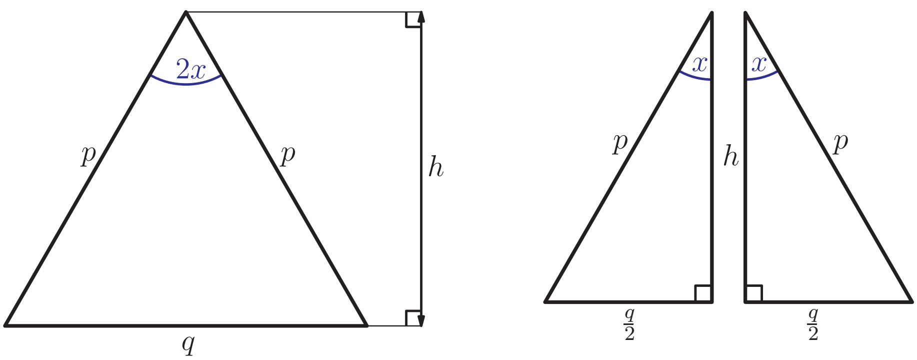
Working in radians, use your calculator to find:

1. $\sin 1.2$ and $\sin 2.4$
2. $\cos 1.2$ and $\cos 2.4$

It seems that there is no connection between the values of trigonometric functions of an angle and those for twice that angle. Are there any rules that relate $\sin 2x$ and $\cos 2x$ to $\sin x$ and $\cos x$?

To try and find such rules, a sensible starting point would seem be the familiar right-angled triangle containing the angle x. We are interested in the angle $2x$ which we can make by simply adjoining an identical right-angled triangle as shown.

We used this idea of joining two right-angled triangles to derive sine and cosine rules in chapter 11.



Let us consider the whole isosceles triangle, and find its area using two familiar formulae.

Using the formula Area $= \frac{1}{2}ab\sin\hat{C}$ we get:

$$\text{Area} = \frac{1}{2}p^2\sin(2x) \qquad (1)$$

We can also calculate the area by using the length of the base and the height of the triangle and the formula $\text{Area} = \frac{1}{2}bh$.

Working in one of the right-angled triangles we have:
$h = p\cos x$

and $$\frac{q}{2} = p\sin x \quad \Rightarrow \quad q = 2p\sin x$$

So a second expression for the area of the triangle is:

$$\text{Area} = \frac{1}{2}(2p\sin x)(p\cos x) = p^2\sin x\cos x \qquad (2)$$

Comparing the two expressions (1) and (2) for the area, we get:

$$\frac{1}{2}p^2\sin(2x) = p^2\sin x\cos x$$

EXAM HINT

You will often see sin, cos and tan of a multiple of x written without brackets: sin2x, cos5x, etc.

Rearranging this equation gives the **double angle identity** for sine.

KEY POINT 12.1

$$\sin(2x) = 2\sin x\cos x$$

Although x was assumed to be an acute angle in the derivation of Key point 12.1, the identity actually applies to all values of x.

Working from the same triangle and using the cosine rule we can derive another double angle identity for cosine:

$$\cos(2x) = \cos^2 x - \sin^2 x$$

See Fill in proofs 9–11 on the CD-ROM.

Substituting $\sin^2 x = 1 - \cos^2 x$ or $\cos^2 x = 1 - \sin^2 x$ gives us two further ways of expressing this double angle identity.

KEY POINT 12.2

$$\cos 2x = \begin{cases} 2\cos^2 x - 1 \\ 1 - 2\sin^2 x \\ \cos^2 x - \sin^2 x \end{cases}$$

A useful application of Key point 12.2 is finding the exact values of half-angles.

The really useful applications of double angle identities are in proving more complex identities and in solving equations.

Worked example 12.1

Using the exact value of $\cos 30°$, show that $\sin 15° = \sqrt{\dfrac{2-\sqrt{3}}{4}}$.

We know that $\cos 30° = \dfrac{\sqrt{3}}{2}$ but we need a way of relating this to $\sin 15°$
Since $30 = 2\times 15$, the obvious choice seems to be the cosine double angle identity

Using $\cos(2x) = 1 - 2\sin^2 x$

continued . . .

$$\cos(2\times 15°) = 1 - 2\sin^2 15°$$
$$\Rightarrow \quad \cos 30° = 1 - 2\sin^2 15°$$

As $\cos 30° = \frac{\sqrt{3}}{2}$

$$\frac{\sqrt{3}}{2} = 1 - 2\sin^2 15°$$
$$\Rightarrow \quad \sqrt{3} = 2 - 4\sin^2 15°$$
$$\Rightarrow \quad 4\sin^2 15° = 2 - \sqrt{3}$$
$$\Rightarrow \quad \sin^2 15° = \frac{2-\sqrt{3}}{4}$$

We have to choose between the positive and negative square root here. $\sin 15° > 0$ so we take the positive square root

$$\therefore \sin 15° = \sqrt{\frac{2-\sqrt{3}}{4}} \qquad (\text{as } \sin 15° > 0)$$

Although they are called the double angle identities, these identities can also be used with higher multiples.

In chapter 19 we shall use double angle identities to integrate some trigonometric functions.

Worked example 12.2

Find an expression for $\cos 4x$ in terms of:

(a) $\cos 2x$ (b) $\cos x$

$4x = 2\times(2x)$, so use one of the cosine double angle identities. Since we want an expression involving only cos it has to be $\cos(2a) = 2\cos^2 a - 1$ with $a = 2x$

(a) $\cos 2(2x) = 2\cos^2(2x) - 1$

$\Rightarrow \cos(4x) = 2\cos^2(2x) - 1$

Use the answer from the previous part

(b) From part (a)

$\cos(4x) = 2\cos^2(2x) - 1$

Replace $\cos(2x)$ in the answer to part (a) with an expression involving only $\cos x$

$= 2(2\cos^2 x - 1)^2 - 1$

EXAM HINT

In the exam, any equivalent form of the answer would be acceptable. Note that, unless you are asked to do so, there is no need to go any further than we did for part (b), for example by expanding the brackets.

Recognising the form of double angle formulae can be useful in solving trigonometric equations.

Worked example 12.3

Solve the equation $6\sin x\cos x = 1$ for $-\pi < x < \pi$.

We see the LHS is similar to 2sinx cosx (which equals sin 2x), so isolate this and use the double angle identity

$$6\sin x\cos x = 1$$

$$\Rightarrow 2\sin x\cos x = \frac{1}{3}$$

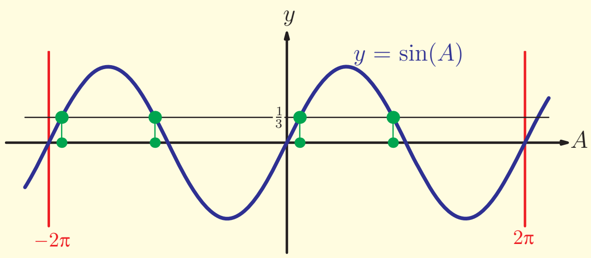
$$\Rightarrow \sin(2x) = \frac{1}{3}$$

Follow the standard procedure and look at the graph to find that there are 4 solutions in the given domain

Substitute $A = 2x$

Since $-\pi < x < \pi$

$-2\pi < 2x < 2\pi$



$$\arcsin\left(\frac{1}{3}\right) = 0.3398$$

$\therefore A = 0.3398, 2.802, -5.943, -3.481$

$\therefore x = 0.170, 1.40, -2.97, -1.74$ (3SF)

If an equation contains both a $\cos 2\theta$ and a $\cos\theta$, we can use identities to turn it into an equation involving only $\cos\theta$.

Worked example 12.4

Solve the equation $\cos 2x = \cos x$ for $0° \le x \le 360°$.

$\cos(2x) = \cos x$

As before, write the equation in terms of only one trig function (Note that $\cos(2x)$ and $\cos x$ are not the same function!)

We can write the LHS in terms of cosine using $\cos 2x = 2\cos^2 x - 1$

$\Rightarrow 2\cos^2 x - 1 = \cos x$

Rewrite as a quadratic in $\cos x$ and factorise

$\Rightarrow \quad 2\cos^2 x - \cos x - 1 = 0$

$\Rightarrow \quad (2\cos x + 1)(\cos x - 1) = 0$

$\Rightarrow \quad \cos x = -\frac{1}{2}$ or $\cos x = 1$

Solve each equation separately

When $\cos x = -\frac{1}{2}$, $x = 120°$ or $360° - 120° = 240°$

When $\cos x = 1$, $x = 0°$ or $360°$

List all the solutions

$x = 0, 120°, 240°, 360°$

There is also a double angle identity for the tangent function.

KEY POINT 12.3

$$\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$$

You should be able to prove this.

Worked example 12.5

Prove that $\tan 2x = \dfrac{2\tan x}{1-\tan^2 x}$.

We know the formulae for sin 2x and cos 2x, so write tan 2x in terms of sin(2x) and cos(2x)

$$\tan(2x) = \frac{\sin 2x}{\cos 2x}$$

Convert from double to single angles using the sine and cosine identities

Use $\cos(2x) = \cos^2 x - \sin^2 x$

$$= \frac{2\sin x \cos x}{\cos^2 x - \sin^2 x}$$

Divide through by $\cos^2 x$ to leave $1 - \tan^2 x$ in the denominator

$$= \frac{2\left(\frac{\sin x}{\cos x}\right)}{1 - \frac{\sin^2 x}{\cos^2 x}}$$

$$\therefore \tan(2x) = \frac{2\tan x}{1-\tan^2 x}$$

In Worked example 12.5 the most useful version of the cos(2x) identity may not have been obvious. The good news is that here, and in many other cases, even if you choose the 'wrong' one you can still complete the calculation; it may just take a little longer.

Exercise 12A

1. (a) (i) Given that $\cos\theta = -\frac{1}{4}$, find the exact value of $\cos 2\theta$.

   (ii) Given that $\sin A = -\frac{2}{3}$, find the exact value of $\cos 2A$.

   (b) (i) Given that $\sin x = \frac{1}{3}$ and $0 < x < \frac{\pi}{2}$, find the exact value of $\cos x$.

   (ii) Given that $\sin x = \frac{3}{5}$ and $0 < x < \frac{\pi}{2}$, find the exact value of $\cos x$.

   (c) (i) Given that $\sin x = \frac{1}{3}$ and $0 < x < \frac{\pi}{2}$, find the exact value of $\sin 2x$.

   (ii) Given that $\sin x = \frac{3}{5}$ and $0 < x < \frac{\pi}{2}$, find the exact value of $\sin 2x$.

2. Find the exact value of:

   (a) $\sin^2 22.5°$ (b) $\cos^2 75°$ (c) $\cos^2\left(\frac{\pi}{12}\right)$

3. Find the exact value of $\tan 22.5°$.

4. Simplify using a double angle identity:

(a) $2\cos^2(3A)-1$ (b) $4\sin 5x\cos 5x$

(c) $3-6\sin^2\left(\frac{b}{2}\right)$ (d) $5\sin\left(\frac{x}{3}\right)\cos\left(\frac{x}{3}\right)$

5. Use an algebraic method to solve these equations:

(a) $\sin 2x = 3\sin x$ for $x \in [0, 2\pi]$

(b) $\cos 2x - \sin^2 x = -2$ for $0° \le x \le 180°$

(c) $5\sin 2x = 3\cos x$ for $-\pi < x < \pi$

(d) $\tan 2x - \tan x = 0$ for $0° \le x \le 360°$

6. Prove these identities:

(a) $(\sin x + \cos x)^2 = 1 + \sin 2x$

(b) $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(c) $\tan 2A - \tan A = \dfrac{\tan A}{\cos 2A}$

(d) $\tan\alpha - \dfrac{1}{\tan\alpha} = -\dfrac{2}{\tan 2\alpha}$

7. Find all the values of $\theta \in [-\pi, \pi]$ which satisfy the equation $\cos^2\theta + \cos 2\theta = 0$. *[6 marks]*

8. Show that $\dfrac{1-\cos 2\theta}{1+\cos 2\theta} = \tan^2\theta$. *[5 marks]*

9. (a) Given that $\tan\alpha\tan 2\alpha = 6$, find the possible values of $\tan\alpha$.

(b) Solve the equation $\tan\alpha\tan 2\alpha = 1$ for $\alpha \in (0, \pi)$, giving your answer in terms of π. *[6 marks]*

10. Express $\cos 4\theta$ in terms of:

(a) $\cos\theta$ (b) $\sin\theta$ *[6 marks]*

In chapter 15, you will learn an alternative method for expressing $\cos n\theta$ in terms of $\cos\theta$ and $\sin\theta$.

11. (a) Show:

(i) $\cos^2\left(\frac{1}{2}x\right) = \frac{1}{2}(1+\cos x)$

(ii) $\sin^2\left(\frac{1}{2}x\right) = \frac{1}{2}(1-\cos x)$

(b) Express $\tan^2\left(\frac{1}{2}x\right)$ in terms of $\cos x$. *[8 marks]*

12. Given that $a\sin 4x = b\sin 2x$ and $0 < x < \frac{\pi}{2}$, express $\sin^2 x$ in terms of a and b. *[6 marks]*

12B Compound angle identities

In Section 12A we derived identities for trigonometric functions of double angles. We now look at identities for expressions like $\sin(x+y)$. These are often called the **compound angle identities**.

Using a generalisation of the methods for the sine and cosine double angle identities from Section 12A we can show that:

See Fill-in proofs 10 'Sine compound angle formula' and 11 'Cosine compound angle formula' on the CD-ROM.

KEY POINT 12.4

Compound angle identities

$$\sin(x+y) = \sin x\cos y + \cos x\sin y$$
$$\sin(x-y) = \sin x\cos y - \cos x\sin y$$
$$\cos(x+y) = \cos x\cos y - \sin x\sin y$$
$$\cos(x-y) = \cos x\cos y + \sin x\sin y$$

EXAM HINT

In the Formula booklet these identities are summarised as

$$\sin(x \pm y) = \sin x\cos y \pm \cos x\sin y$$
$$\cos(x \pm y) = \cos x\cos y \mp \sin x\sin y$$

Notice the signs in the cosine identities: In the identity for the sum we use the *minus* sign, and in the identity for the difference we use the *plus* sign.

Note that the compound angle formulae can be used to derive the double angle formulae by setting $x = y$.

As with the double angle identities, one of the simplest applications is to calculate exact values of trigonometric functions.

Worked example 12.6

Find the exact values of:

(a) $\sin 75°$ (b) $\cos 15°$

We know the exact values of sin for a few angles: $0°, 30°, 45°, 60°, 90°$

Use $30° + 45° = 75°$

(a) $\sin 75° = \sin(30° + 45°)$

Use the sine addition identity

$= \sin 30° \cos 45° + \cos 30° \sin 45°$

$= \frac{1}{2}\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\frac{\sqrt{2}}{2}$

$= \frac{\sqrt{2}+\sqrt{6}}{4}$

$15° = 45° - 30°$ so use cosine subtraction identity

(b) $\cos 15° = \cos(45° - 30°)$

$= \cos 45° \cos 30° + \sin 45° \sin 30°$

$= \frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\frac{1}{2}$

$= \frac{\sqrt{6}+\sqrt{2}}{4}$

We can now use the sine and cosine compound angle identities to derive new ones. In particular:

KEY POINT 12.5

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Again, you should be able to prove this.

Worked example 12.7

Prove that $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$.

Express tangent in terms of sine and cosine.

$$\tan(x + y) = \frac{\sin(x + y)}{\cos(x + y)}$$